Media and Political Competition with Rationally Inattentive Voters*

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Abstract

We study how the rise of Social Media affects professional news quality, media profits, voter attention, and political belief polarization. Traditional Media produces information about a common policy-relevant state, while Social Media intermediates this content and combines it with personalized information. Voters are rationally inattentive and choose how much additional costly information to acquire. We show that greater Social Media penetration can weaken incentives to invest in professional reporting when the returns from intermediated audiences are sufficiently low. In our baseline calibration, this reduces both reporting quality and Traditional Media's profits. Revenue sharing introduces a strategic trade-off: it encourages professional reporting but weakens Social Media's incentives to transmit that information faithfully. These decisions shape voter attention and the dispersion of political beliefs through personalized information and differences in the quality of common information. Our numerical results illustrate how their interaction can generate non-monotonic effects on reporting quality and belief polarization.

Keywords: Media, Rational Inattention, Information Production, Voting, Belief Polarization.

JEL Classification: D83, D82, L82, D72.

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1 Introduction

Digital platforms have transformed the way political information reaches voters. Professional news outlets continue to undertake the costly work of producing original reporting, but an increasing share of that information is encountered through intermediaries that aggregate, rank, summarize, and distribute content. These platforms also collect information generated by users' activity and use it to personalize what individuals see. The resulting information environment combines a common source of professional reporting with potentially different information for each voter. The distinction between information production and intermediation is central to the economics of Social Media (Aridor et al., 2024) and to the literature on news aggregation and publishers' incentives (Dellarocas et al., 2013; Jeon and Nasr, 2016; Chiou and Tucker, 2017; Athey et al., 2021). It raises a question that extends beyond how much news is available: how do the incentives to produce and transmit information determine what voters ultimately learn?

This paper studies the interaction between professional reporting, platform curation, and costly voter attention. We ask how the growth of Social Media and the division of platform revenues with an original news producer affect the quality of common information, the acquisition of personalized information, and the distribution of voters' posterior expected policy positions. Our focus is not on a platform introducing an ideological bias into a report. Rather, even when signals are unbiased, decisions about their precision and transmission can change both the accuracy and the dispersion of political beliefs. We distinguish this belief-based notion of polarization from the exogenous distance between political parties' platforms, which determines the stakes of becoming better informed.

We develop a sequential model with a Traditional Media information producer, a Social Media intermediary, and a continuum of rationally inattentive voters. Each voter's unknown preferred policy position is the sum of a common component and an idiosyncratic component. Traditional Media invests in the precision of a report about the common component. A fraction of voters receive this report directly; the remaining, exogenously assigned fraction obtain news through Social Media. After observing the reporting decision, the platform chooses its transmission quality, which determines how much precision of the original professional report reaches its users. In addition, Social Media supplies personalized information about individual preferences. The precision of this personalized signal increases with the size of its user base, capturing in reduced form the value of aggregated social data (Bergemann et al., 2022).

Voters initially receive costless information from their assigned platform and subsequently decide how much additional costly information to acquire. The distinction between media

technologies remains relevant in this second stage: a Traditional Media viewer can learn further about the common state but cannot learn the idiosyncratic component through that outlet, whereas a Social Media viewer can acquire information about the entire preferred policy position. Information acquisition is therefore shaped jointly by the quality of the first-stage signals and the dimensions of preferences about which additional learning is feasible. Consistent with rational inattention (Sims, 2003; Maćkowiak et al., 2023) and its political applications (Matějka and Tabellini, 2021), voters invest more in learning when the stakes of the electoral decision are greater. A more informative initial report, however, reduces the residual uncertainty and can crowd out subsequent costly attention. This interaction between information supplied by a sender and information acquired by a receiver is related to Matysková and Montes (2023), while the commercial importance of attention connects the model to Galperti and Trevino (2020) and the quality-attention trade-off discussed by Serra-Garcia (2026).

Media outlets anticipate learning responses from voters. Traditional Media benefits from the commercial value of reporting quality and direct-reader attention, but must pay the cost of producing a more precise report. Social Media values the effective quality of professional information shown to its users and their subsequent attention, while incurring a cost of high-fidelity transmission. A predetermined revenue share, λ, specifies how much of the platform's gross revenue is transferred to Traditional Media. We interpret this parameter as a reduced-form feature of publisher-platform arrangements rather than the outcome of an explicitly modeled bargaining problem. In equilibrium, the news producer chooses reporting precision first, the intermediary chooses transmission quality second, and voters then make their information-acquisition decisions. This vertical relationship distinguishes the analysis from models focused exclusively on competition among original news producers or on the bias of their reports.

Our first set of results concerns the production of professional information. In the Traditional Media monopoly benchmark, more precise initial reporting reduces the incentive to acquire additional information; for sufficiently high political stakes, the restriction on learning about idiosyncratic preferences becomes binding. With platform entry and exogenously fixed transmission quality, reporting incentives also depend on the relative marginal returns generated by direct and intermediated viewers. When the revenue-share marginal return from an additional platform viewer is sufficiently low, an increase in Social Media penetration reduces equilibrium reporting precision. This is a conditional result, not a mechanical consequence of audience migration: the way value generated through the platform is shared with the original producer matters. This mechanism complements the literature examining how aggregation changes publishers' incentives and the returns to directing users to original

news sources (Dellarocas et al., 2013; Jeon and Nasr, 2016).

Allowing Social Media to choose transmission quality introduces a further strategic margin. More faithful transmission increases the commercial value of professional information on the platform, but can also reduce subsequent costly attention and entails a transmission cost. Increasing λ raises the fraction of platform revenues received by Traditional Media while reducing the revenues retained by Social Media. The platform therefore has weaker incentives to preserve professional reporting, and the producer anticipates this response. In our numerical equilibrium, transmission quality declines with the revenue share, while reporting precision first rises and then falls. This hump-shaped relationship contrasts with the positive reporting response to revenue sharing in the fixed-transmission comparison and illustrates why compensation rules cannot be evaluated independently of the intermediary's curation incentives. We do not claim that the hump shape is universal; it is an equilibrium outcome of the illustrated parameterization.

We next connect these market incentives to political belief dispersion (as a measure for voter's polarization). Our measure is the expected conditional variance of voters' final posterior expected policy positions, conditioning on the original professional report. It decomposes into the weighted dispersion within the Traditional and Social Media groups and a term reflecting differences between their conditional average beliefs. This distinction matters because the platform's imperfect transmission of a common report can generate a gap between the groups' conditional means even when signals are unbiased. Personalization provides another source of dispersion by revealing differences in voters' idiosyncratic preferences. Yet this force is moderated by endogenous learning: when voters receive more informative costless signals, they may acquire less additional information. Hence Social Media does not necessarily produce a strictly more dispersed distribution of final posterior means for every information environment. Under faithful transmission, equal attention costs, and interior learning, the distributions can coincide; at sufficiently high political stakes, the limitation on Traditional Media learning instead produces a dispersion gap associated with the variance of idiosyncratic preferences.

Hence, greater dispersion of posterior expectations can reflect more accurate learning about heterogeneous underlying preferences rather than a deterioration in voters' information. Moreover, because we condition on the original report rather than on every realized common signal, the statistic is an expected conditional distribution of beliefs, not necessarily the realized disagreement in one electorate after all common information has been fixed. Our analysis therefore concerns the polarization of learned policy positions, not endogenous party-platform divergence, electoral outcomes, or a welfare ranking of different information environments. This interpretation connects to Vaeth (2026), who studies how rational

learning about heterogeneous ideal points can generate dispersion in revealed ideology. Our contribution is to endogenize the professional reporting and platform transmission decisions that determine the information from which those beliefs are formed.

Finally, we study how revenue sharing shapes this polarization measure in equilibrium. In our example, aggregate conditional belief dispersion initially falls with λ because the improvement in professional reporting more than offsets a deterioration in transmission quality. The minimum occurs before reporting precision reaches its maximum: transmission continues to weaken even while reporting is still improving. At higher revenue shares, the transmission effect dominates and, once reporting also begins to decline, the two forces reinforce one another. Counterfactual calculations that vary reporting and transmission separately show that the transmission channel is quantitatively important in the illustrated equilibrium. Thus, the revenue-sharing arrangement that maximizes professional reporting need not minimize the dispersion of voters' final posterior beliefs. The result highlights a distinction between incentives to produce original news and incentives to deliver it faithfully.

The way we will proceed in this paper is by discussing briefly the literature related to our approach in Section 2. Section 3 presents some empirical evidence. Section 4 presents a general description of the model and the timing of the game. Section 5 characterizes the Traditional Media monopoly benchmark, while Section 6 studies the equilibrium with Social Media, first with fixed and then with endogenous transmission quality. Section 7 derives the conditional belief-dispersion measure and examines how it responds to the media-market equilibrium. Finally, section 8 concludes and discuss several future extensions.

2 Literature Review

This paper contributes to the literature on how digital platforms reshape the production and distribution of political information. As emphasized by Aridor et al. (2024), Social Media differs from traditional news outlets in its role as an intermediary that combines content produced by third parties with user-generated information. A growing literature examines how news aggregation affects the consumption of original news and the incentives of publishers (Dellarocas et al. (2013); Chiou and Tucker (2017); Jeon and Nasr (2016); Athey et al. (2021)). Particularly relevant is the work of de Cornière and Sarvary (2023), who study how the bundling of professional news and social content affects publishers' incentives to invest in reporting quality. Their analysis highlights how platforms can change the returns to quality investment, with effects that depend on publishers' characteristics and the arrangements governing content distribution.

Our paper builds on this literature by distinguishing two decisions that jointly determine

the quality of information received by voters: the precision of the original professional report and the fidelity with which the platform transmits that information. We consider a single original news producer and a Social Media intermediary, rather than competition among multiple news producers, and take the division of platform revenues as given. This allows us to examine how revenue sharing affects not only the original producer's incentives to invest in reporting, but also the intermediary's incentives to preserve the informativeness of that report. The distinction is consequential because increasing the producer's share of revenues can improve reporting incentives while weakening transmission incentives, giving rise to potentially non-monotonic equilibrium effects.

A second strand of related work examines the interaction between information supply and consumers' costly attention. Chen and Suen (2023) show how competition for attention can weaken news providers' incentives to invest in quality, as consumers allocate their attention across competing sources. Galperti and Trevino (2020) study endogenous information accuracy and clarity in markets where information providers monetize consumers' attention, emphasizing the importance of coordination motives. Our framework shares the idea that information quality and attention are jointly determined, but focuses on a vertically related producer and intermediary rather than competing original information suppliers. An additional connection is Matysková and Montes (2023), who analyze how information provided by a sender interacts with a receiver's incentive to acquire independent information at a cost. In our model, more informative initial signals similarly reduce the residual uncertainty that voters can resolve through subsequent costly learning. This creates a trade-off for media firms between the commercial value of more informative news and the attention it may displace. Serra-Garcia (2026) provides complementary experimental evidence that incentives to attract attention can affect how much information is transmitted to consumers.

Our distinction between the two media technologies also relates to the economics of social data (Bergemann et al., 2022). Social Media not only intermediates professional reporting about a common policy-relevant state, but also provides personalized information about voters' idiosyncratic preferences. We capture the informational advantage associated with a larger user base through a reduced-form relationship between platform penetration and the precision of personalized signals. The resulting information environment combines shared professional news with individualized information whose informativeness varies with the platform's reach.

Finally, our paper contributes to the literature on rational attention and political belief formation. Matějka and Tabellini (2021) show how voters allocate costly attention in response to the stakes of electoral competition, with consequences for political incentives and policy choices. More closely related to our analysis of belief dispersion, Vaeth (2026)

demonstrates that rational learning about heterogeneous individual preferences can itself generate polarized revealed ideology, even without biased information or media influence. Our approach differs by endogenizing the market decisions that determine the information initially available to voters and the dimensions of their preferences about which additional learning is feasible. We take political parties' positions as given and examine how professional reporting, platform transmission, and personalized information shape the distribution of voters' posterior expected ideal points.

This focus also distinguishes our analysis from conventional models of media bias, in which information is strategically slanted or selectively suppressed (Gentzkow and Shapiro, 2006), and from models in which media competition and issue selection polarize voters' perceptions of issue importance (Yamaguchi, 2022). In our framework, signals need not exhibit ideological bias for the information environment to affect political belief dispersion. Imperfect transmission can create differences in conditional average beliefs across media audiences, while personalized information and endogenous learning shape the dispersion of beliefs within each audience. Consequently, greater dispersion may reflect either differences in access to common information or more accurate learning about genuinely heterogeneous preferences. Our contribution is to connect these informational mechanisms to the equilibrium incentives of a professional news producer and a Social Media intermediary.

3 Empirical Motivation

3.1 Data

To motivate our theoretical model we rely on data from the *European Social Survey*. This is a biannual survey, starting from the year 2002, measuring a wide range of societal attitudes and opinions across European countries, with the latest wave of the study being released for the year 2022, and including up to 24 countries[1]. Crucially, this study does not represent a continuous panel, but instead a new sample is taken for each country every time a new interview wave is conducted. For this reason, we end up working with a repeated cross-section.

During each wave, the selected participants are surveyed on their opinions about a wide range of political, societal, environmental and cultural topics. Moreover, it zooms into the political preferences of participants, while at the same time provides a glimpse into their

[1]The sample of countries varies from wave to wave. For this reason sometimes we will end up having more data from some countries than others, while the number of countries in the total sample will be larger than the 24 that appeared in the last wave, published in 2024.

habits and, of interest to our study, their consumption and engagement with news pertaining politics and current issues.

Given that we are interested mainly in the impact that different media sources may have on voters' ideology and, thus, on political competition, we will focus on the variables in the study that report the time a person spends getting news about current issues or politics from a media outlet. Moreover, we will contrast this with measures of self-reported interest in politics and, for more recent waves, on self-reported online content sharing about politics-relevant issues.

3.2 Descriptive evidence

The wide time-span of almost 20 years the survey covers allows us to identify some patterns across groups of voters in the economy, pertaining mostly to their use of media when focusing on topics that might be of interest when choosing one candidate over the other during elections. That is, we abstain from tracking overall media consumption and constraint our attention to self-reported consumption of news on politics or current affairs.

According from reports of the *Reuters Institute* (Newman et al. 2025) the use of social media a channel through which people get their news from has seen a rapid increase in the last decade all over the world. In countries like the United States and Brazil, almost one-third of people cite social media as their main source for getting news from.

Moreover, there is wide variability of usage of different media sources across age groups. The *Reuters Institute's* report finds that in 2025 around 45% of people under 25 years of age get their news mainly from social media. On the other hand, more than half of the people over 55 go to more traditional media sources such as TV, newspapers or radio to inform themselves.

When going to the more granular data available in the *European Social Survey*, an inspection of the data on the amount of minutes spent following political/current issues news, and online-sharing activity leaves us with two important insights. Firstly, as people grow older they tend to invest more time (a proxy for attention) into following political news, however the youngest generations are those that share more political content online, when focusing on the subsample between the years 2016 and 2022.

This potentially makes that the news that people get from social media are shaped by content shared by people that tend to pay less attention to news. Hence, if media is important in shaping the views of people, this may create a bias in the content that is being shared online. In turn, passive users of social media consume content that, although catered to them in some degree, feeds (or is distorted) by the content produced by users from a more

active group[2]

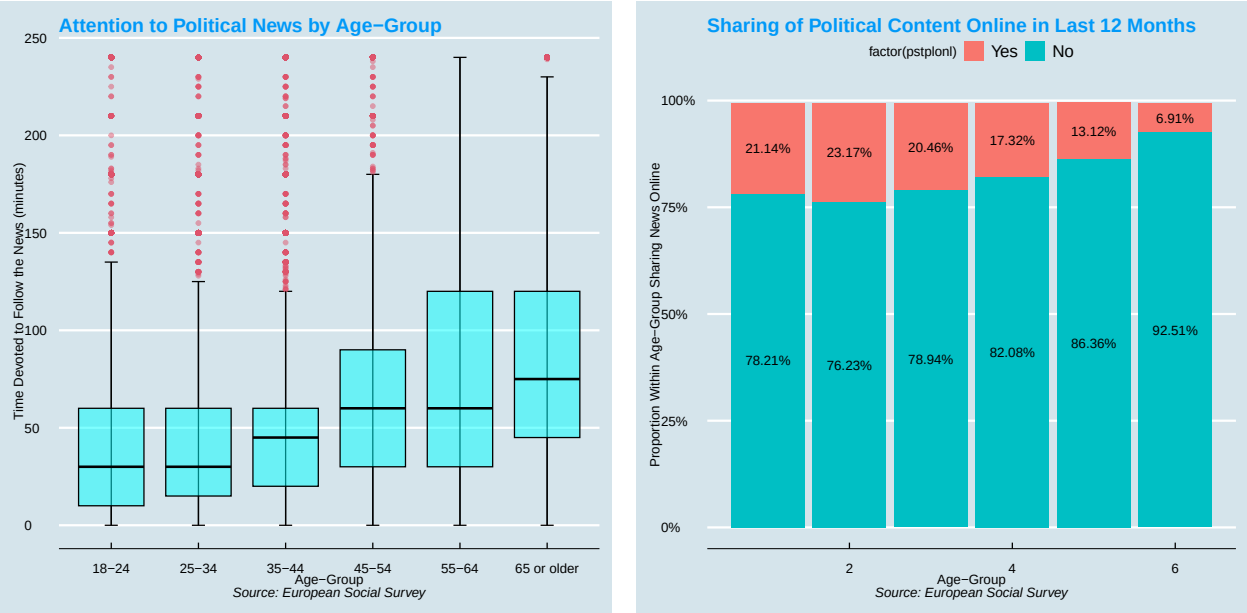


Figure 1: Attention to political news by age-group (left) and sharing of political content online divided by age (right)

Since we are ultimately interested in the polarizing role that media may have on voters' views and on political parties, we complement the European Social Survey with a variable that indicates if a person voted in the last election for a party that is considered either extremist or populist.

We do this to control for a potential source of heterogeneity between people that vote for mainstream parties and those that cater towards more unconventional, or polarizing, ones.

To construct this variable we resorted to the documentation provided by the European Social Survey itself, and choose those parties that were described either as *far-left/right* or *populist*. We contrasted this information with the left-right score constructed by the *Chapel Hill Expert Survey*. The latter is a qualitative study in which a panel of country experts assign a value between 0 and 10 to a party, so as to position it within a *left-right* scale. We assume that a party is *extremist* if it has a score lower than 1.5 or higher than 8.5, with values closer to 0 indicating that a party is considered *far-left*, while values closer to 10 signal that such a party is considered *far-right*.[3]

[2]In this case, we think of *active* as defining a group of people that shares political content online with more frequency, or intensity.

[3]In this sense, we take an *ad-hoc* cutoff to separate parties. However, the parties that fall under these *extremist* categories are widely considered as such by a myriad of alternative sources. Importantly, it might be more accurate to refer to our variable as encompassing voting for *outsider* parties, as it includes political parties that are not extremist in their policy views, but rely on a *populist* way of doing politics (i.e Cinque

3.3 Regression Results

We conducted two exploratory regressions to see what tends to correlate with people paying more or less attention to political news, and what shapes their positioning on the left-right political spectrum. Both regressions take into account analytical weights and robust standard errors. Moreover we control for time and country dummies.

Table 1: OLS regressions on attention to news and position on Left-Right scale

Variable	Attention to News	Left-Right Scale
Extremist/Populist Vote	9.5052***	-
	(3.567)	-
Age Group	6.3764***	0.1441***
	(0.5109)	(0.0059)
Interest in Politics	14.6936***	0.1053***
	(0.9643)	(0.0119)
Worked at Political Org.	6.5185*	-0.4243***
	(3.6176)	(0.0876)
Country Dummy	Yes	Yes
Year Dummy	Yes	Yes

Standard errors in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Our main regressions suggest that those that vote for extremist or populist parties, according to our definition, tend to pay higher attention to news. This seems to be in line with the fact that more polarized voters tend to pay higher attention to policy issues (Matejka and Tabellini. 2021). Moreover, we see that age is indeed a factor in explaining differences in attention to news, as figure 1 suggested.

Now we turn our attention to a *probit* model to disentangle which factors appear to affect the decision of share political content online, where for the sake of comparison we employ the same control variables as in our OLS framework.

The results of our probit model, displayed in table 3, tells us that more polarized voters tend to share more political content online. Also, as expected, as people grow older they have a lower propensity to share information on the internet. We also control in other specifications for other characteristics and set of values. Particularly, we also find that people have a higher propensity to share content online when they report to be less satisfied with the government.

Finally, one important question when talking about political competition is to get a sense what makes voters more prone to vote. Although in our theoretical model all agents *do* cast

Stelle in Italy)

Table 2: Probability of sharing political content online

Variable	Share Political Content
Extremist/Populist Vote	0.2104***
	(0.125)
Age Group	-0.3149***
	(0.021)
Interest in Politics	0.09398***
	(0.0031)
Worked at Political Org.	0.8907***
	(0.0212)
Country Dummy	Yes
Year Dummy	Yes

Standard errors in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

a vote, inspecting this question is relevant when thinking about the potential implications for the incentive of political parties and media when conveying information to voters.

Basically, the evidence tells us that, after controlling for country dummies, the probability that a person do goes increases if you posted or shared political content online recently. Suggesting that more engaged voters are more prone to cast their votes on election day, something supported by self-reported interest in politics.

Table 3: Probability of voting

Variable	Vote
Share Political Content	0.1709***
	(0.0120)
Age Group	0.1371***
	(0.0024)
Interest in Politics	0.4148***
	(0.0048)
Trust in Politicians	0.0291***
	(0.0020)
Trust in Legal System	0.0324***
	(0.0018)
Country Dummy	Yes

Standard errors in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Interestingly, as people age their propensity to vote increases. This could suggest that a non-traditional outlet (such as social media), interested on its impact on voters, faces a potential trade-off because as voters age they share less political content (meaning they use the platform less), but they do vote with a greater propensity.

4 Theoretical Model

Consider an economy populated by a continuum of unit mass of households, $i \in \mathcal{I}$, who need to learn about their underlying preferences (form an opinion) about a policy-relevant issue to cast the vote in an election between two political parties, $j = \{L, R\}$. Call this underlying preferences the voter's **policy stance**. We denote this by θ_i.

We will be more precise about this in the main section, but we will assume the voters hold a prior belief on what their own θ is. To refine knowledge about this they may consume information (i.e. acquire signals) from two sources k: Traditional Media and Social Media. We will denote the former with $k = T$, whereas Social Media will be identified with $k = S$.

In reality, Traditional Media outlets and Social Media are structurally different business models. We can think of traditional outlets as being information producers, whereas social media acts as an intermediary of content between traditional outlets and information consumers. This makes for a two-sided-market-like setting in which the social media platform balances what two sides of one market bring to the interaction.

Another realistic feature is that social media does not only encompasses content generated by traditional outlets, but they also rely on *user generated content*. Think of the latter as being comprised of any post that is shared or created by any person in the social media's ecosystem. It is this second source of information the one that allows the platform to get information about voter-specific characteristic outside the reach of Traditional Media.

Traditional Media chooses the precision $\psi \geq 0$ of its professional reporting. Social Media subsequently chooses $\omega \in [0, 1]$, the fidelity with which this reporting is transmitted through the platform feed. In particular, when the original report has precision ψ, the professional component of the signal received by a Social Media user has effective precision $\omega\psi$. At $\omega = 1$, the original report is transmitted without additional distortion; at $\omega = 0$, it conveys no information about the common state. Thus, ω represents *effective informational fidelity*, rather than a literal fraction of stories displayed. The transfer rule that allocates a fraction λ of Social Media revenues to Traditional Media is taken as given by both firms.

Voters are assumed to be Rationally Inattentive (Sims (2003), Maćkowiak et al. (2023)) in their information processing. In short, this means they have some constraint as to how much information they can process. However, every information bit of information they

process will be done in a rational way. In our context, this means the voters will update their priors on θ upon receiving a signal from the chosen media outlet in a standard Bayesian way.

As standard in the Rational Inattention literature with Mutual Information costs (see Matějka and Tabellini (2021) for an example directly applied to political competition), the value of costly information acquisition is directly related to the prior uncertainty the agent has. Thus, we adopt a framework in which learning about θ is done in two stages, so as to allow media decisions to influence the voter's prior information. More precisely, we assume that the media would send a first costless signal (from the voter's point of view) that will allow the latter to update her initial beliefs. In a second stage, the voter can exert costly information acquisition to further refine her knowledge about θ.

The distinction between passive exposure and active research is particularly important for Social Media. A user may respond to the initial feed by opening posts, comparing accounts or videos, checking claims, and following additional content. We model this research as a separate signal whose precision is chosen through attention. Social Media's transmission decision ω affects the quality of *initial* information, but does not directly constrain the precision of this subsequent research. It nevertheless changes the incentive to undertake that research by altering interim uncertainty.

After forming their beliefs through this process, the voters would choose between two proposed policies (one for each party) and cast a vote for the one that is closer to her perceived preferences. Elections are held and final payoffs are realized. Note that the media does not care about the ultimate action the voter takes (who she votes for). Rather, we propose a model in which media has their own incentives in capturing voters' information and maximizing profits, which ultimately shape voters' quality of learning and thus the political beliefs they hold. This is the polarization mechanism we pose in this model: one in which a more polarized society is one with more dispersed beliefs.

4.1 Timing

The economy unfolds sequentially within a single period. At the beginning of the game, Nature draws the common and idiosyncratic components of voters'policy stances, μ and ϵ_i. Voters are exogenously sorted between Social Media users, with probability p, and direct Traditional Media viewers, with complementary probability $1-p$. Political parties' positions and the transfer schedule between the platform and Traditional Media are taken as given.

As a first mover, Traditional Media chooses the precision ψ of its professional reporting about the common component μ. At the same time, Social Media users generate and share

content within the platform's ecosystem upon joining. Both the reporting by traditional media and user activity determine the informational environment available to the platform.

After observing Traditional Media's reporting quality and the content available in its ecosystem, Social Media chooses $\omega \in [0, 1]$, the *transmission quality* assigned to Traditional Media content in the platform's feed. The transfer schedule itself is predetermined; hence the platform's only choice at this stage is ω. This relative transmission quality of the Traditional Media's content would affect the voter's learning by reducing the relative importance of professional information in interim updating.

Voters then receive their first, costless information. Direct Traditional Media viewers observe the a signal about the common component, while Social Media users observe a vector containing both a common component signal and personalized information about their idiosyncratic component. After this, voters update their beliefs and form interim posteriors.

Finally, voters choose how much costly attention to allocate to additional information. After processing the costly signal, final beliefs are formed, voting is conducted, and payoffs are realized.

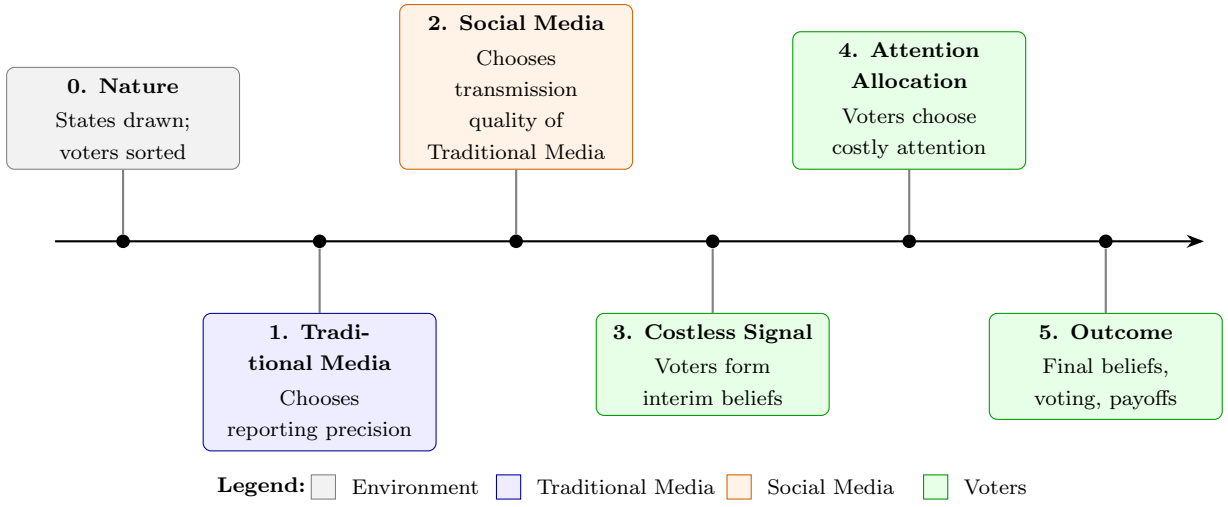


Figure 2: Timing of the media and voter learning game.

4.2 Voters

As mentioned, there is a continuum of mass one of voters, $i \in \mathcal{I}$, that have a policy stance, θ_i, about a sensitive topic for political competition. We can read this as there being a topic that is driving a political campaign in the economy, such as tax policy, inflation, immigration policy, or climate change.

This policy stance will be the sum of a common policy component, μ, and an idiosyncratic

one, ϵ_i. We assume that both of these are normally distributed, such that we can define:

$$\theta_i = \mu + \epsilon_i$$
$$\mu \sim \mathcal{N}\left(m_0, \frac{1}{\tau_\mu}\right) \quad \epsilon \sim \mathcal{N}\left(0, \frac{1}{\tau_e}\right)$$

We have that θ_i is unknown to the voter at the beginning of the game. We assume that the prior is common to all voters, such that before any information is revealed. The mean of the policy preference (i.e. stance) is m_0. For the purpose of illustrating the main results we will work with this zero-prior-mean assumption, and then the prior is $\theta_i \sim \mathcal{N}(0, \sigma_0^2)$.

All the results will be unchanged qualitatively if this is assumed to be different from zero. Furthermore, note that under our assumptions $\sigma_0^2 = \sigma_\mu^2 + \sigma_\epsilon^2$

Voters learn from a media outlet $k = \{T, S\}$ because that would help them learn about which proposed policy will be closest to their preferred stance. As such, the utility of a voter depends on how far is the revealed policy stance from the chosen policy q_j:

$$U_i = -(q_j - \theta_i)^2$$

The learning about θ_i will occur in two stages. First, the household needs to decide from which outlet to acquire a **costless signal** (a headline, or scrolling to the social media's feed), defined as $y_{i,k}$, about the underlying state.

Suppressing for now the sub-index i, since all voters that acquire information form a particular outlet k face the same problem (although with different signal realizations), we have that upon receiving y_k a voter updates her initial prior and forms **interim beliefs** $\hat{\theta}_k$ that depend on the whether she acquired the information from traditional or social media:

$$\hat{\theta}_k \sim \mathcal{N}(m_{1,k}, V_{1,k})$$

We assume in our simplified baseline that outlet adoption is imposed exogenously. Particularly, we set a fraction p of the voter population to getting their news from Social Media, while the complementary fraction $1 - p$ gets their news from traditional sources.

Who is sorted in which outlet matters not only for which information structure is acquired by voters, but also which type of content is generated in Social Media. We assume that the voters that join Social Media will share information about ϵ_i. The nature of this information will be formalized below when we describe both outlets' problems, but this hints towards the fact that Social media users act also as *information producers*.

Upon forming interim beliefs, a voter from outlet k will decide how much costly attention

to allocate to acquire a second signal d_k. This signal can be related to doing some further research on one's own. Likely, this means the voter spends some time reading through one or several articles, or listening for a longer time to the news on TV. In the case of Social media this can be related to fact checking some posts through trusted accounts, or looking at different people's content. We assume the voters cannot go to another outlet in the second stage, leading to the following assumption:

Assumption 1 (Single-Homing)**.** *Once a voter is assigned to an outlet k in the first stage, and acquired a costless signal through it, she can only acquire a costly signal from the same outlet.*

This previous assumption is in line with much of the literature in competition of media sources and platforms; although it can naturally be relaxed and allow for users to be able to contrast two different sources in the second-stage. The structure of the costly signals from each outlet is:

$$d_{i,T} = \mu + e_{i,T} \quad e_{i,T} \sim \mathcal{N}\left(0, \frac{1}{\tau_e}\right)$$
$$d_{i,S} = \mu + \epsilon_i + e_{i,S} \quad e_{i,S} \sim \mathcal{N}\left(0, \frac{1}{\tau_e}\right)$$

Voters are rationally inattentive. In this regard, they have some capacity constraint limiting the amount of attention they can exert to reduce the uncertainty contained in a signal. Once they choose how much effort to invest in attention, the information will be processed in a rational (Bayesian) way.

Following the literature on Rational Inattention (see Sims (2003)), we assume the cost of processing information is **proportional** to the reduction in entropy about a state after paying attention to a signal, d, conditional on having received costless signal $y_{i,k}$:

$$\mathcal{C}_{i,T} = \kappa_k I(\theta_i; d_{i,k} | \mathcal{I}_{1,i,k}) \quad \kappa_k > 0 \tag{4.1}$$

Where κ_k is the media-dependent marginal cost of attention and $I(\cdot)$ is the uncertainty reduced about the policy stance by voter i after allocating attention to signal d_k. This is conditioned in the information set of voter i after receiving the costless signal, $\mathcal{I}_{1,i,k}$

We are assuming that the relevant object for the voter to learn about is with respect to the electoral decision. It does not a direct physical cost of producing an arbitrarily precise observation of whichever component appears in $d_{i,k}$. For Traditional Media viewers, learning about μ also reduces uncertainty about θ_i, but never reveals the idiosyncratic component.

An alternative way to model (4.1) would be to make the learning relevant only to the

elements that d_k contains. Naturally, since Social Media voters can learn about both components of θ_i this does not affect how their learning will be. On the other hand, it does has implications on the optimal learning for Traditional Media voters. If we make this cost proportional only to the uncertainty reduction on μ then reaching the same level of reduction in uncertainty would be more costly to the voter. The interested reader can find a thorough comparison of both approaches in Appendix D. Much of the qualitative conclusions and results remain across both cases.

Note that Traditional Media has in fact a technological constraint, since it cannot report about the idiosyncratic component of voters' policy stances. This creates an effective ceiling on what a voter of T media can learn. Our modeling convention here just makes it more likely than a voter from Traditional Media is more likely to reach a *learning ceiling*. This is a point in which none uncertainty about μ persists, while any uncertainty about ϵ_i is reduced. In the alternative formulation, voters find it prohibitively costly to learn all about μ. In that case, as costly learning increases they approach asymptotically the *learning ceiling* imposed by Traditional Media's technological restriction. This means, that we are adopting the implicit assumption that $d_{i,T}$ can be infinitely precise, as it can reduce all uncertainty about μ.

Our modeling choice of the mutual information cost seeks to exploit the fact that, although Traditional Media voters would have liked to learn as much as possible about θ_i, the information they can actually get is not limited (at least not all of it) by their cognitive capacity, but by a structural difference in the media outlets' transmission technology. This emphasizes the new channels through which Social Media can affect learning, when compared to traditional outlets, by its use of targeted content and its active use of user data.

The signals $d_{i,k}$ characterize the dimensions of the state about which additional information is available through outlet k. The voter does not choose the realization of the signal or its underlying information technology. Following Matějka and Tabellini (2021), we model costly attention as the precision of an additional Gaussian signal chosen by the voter. More precise signals reduce posterior uncertainty but entail higher information-acquisition costs, measured by the reduction in entropy about the policy stance. Nevertheless we add to their framework the fact that endogenous choices

Let $V_{1,k}$ and $V_{2,k}$ denote the interim and final posterior variances of the voter's policy stance θ_i, respectively. We define attention as the fraction of interim uncertainty eliminated through additional learning:

$$\xi_k \equiv 1 - \frac{V_{2,k}}{V_{1,k}}.$$

Importantly, the information that can be acquired differs across media. As mentioned,

Traditional Media enables voters to learn only about the common component μ, leaving uncertainty about ϵ_i irreducible. Consequently, Traditional Media attention is bounded above by

$$\bar{\xi}_T = \frac{a(\psi)}{a(\psi) + b},$$

This is the *learning ceiling* we have referenced above. Here we defined $a(\psi) \equiv 1/(\tau_\mu + \psi)$ and $b \equiv 1/\tau_\epsilon$. We will be more precise on these objects in the following sections, but they represent the reduction in uncertainty due to the common-component report of Traditional Media ($a(\psi)$); and the irreducible uncertainty about ϵ_i (the term b). By contrast, Social Media allows voters to acquire an additional signal about the complete state θ_i, implying $\bar{\xi}_S = 1$.

As noted, we assume that information-processing costs are proportional to the conditional entropy reduction about θ_i upon getting a second costly signal from the same media outlet. Under Gaussian information acquisition, and our stated assumptions, the cost of attention has a form that is not individual-dependent:

$$\mathcal{C}_k(\xi_k) = -\frac{\kappa_k}{2} \log(1 - \xi_k).$$

Thus, we use a common functional form for the information cost, even though the media offer different feasible additional experiments.

This specification extends the entropy-based cost of Matějka and Tabellini (2021) to media-specific information technologies. In particular, it accounts for the fact that Traditional Media voters can eliminate only a fraction of their total uncertainty about θ_i, even with arbitrarily precise additional information.

Optimal attention follows from balancing the expected value of improved information against this cost. Following the local approximation to the value of information in Matějka and Tabellini (2021), the gross benefit is proportional to the reduction in posterior variance, weighted by the political stakes of the voting decision.

Given that optimal learning might be bounded, we define for consistency the set of reducible uncertainty for each media outlet's viewer. Let us call this as Ξ_k, such that for Traditional Media $\Xi_T = [0, \bar{\xi}_T]$, whereas for Social Media $\Xi_S = [0, 1]$. Hence, we can write the costly information allocation problem of the voter of media k as choosing ξ_k within Ξ_k to maximize the following objective:

$$\max_{\xi_k \in \Xi_k} \quad (\Delta u'(q))^2 V_{1,k}(\psi_k)\xi_k + \frac{\kappa_k}{2} \log(1 - \xi_k) \tag{4.2}$$

As mentioned, the objective is a second-order approximation of the utility function of the

voter around the interim beliefs. Here we have called $\Delta u'(q)$ to the (expected) differences in marginal utility of voting for each party j. The derivation of this will be presented in the appendix, and highly resembles the standard optimal attention allocation stage problem found in other papers (see Matějka and Tabellini (2021)). This approximation allows us to write the problem with a focus on the marginal value of costly attention.

Thus, a voter holding interim beliefs $\hat{\theta}_k$ will update her information, upon observing d_k to $\overline{\theta}_k$ such that **final beliefs** follow a normal distribution characterized by:

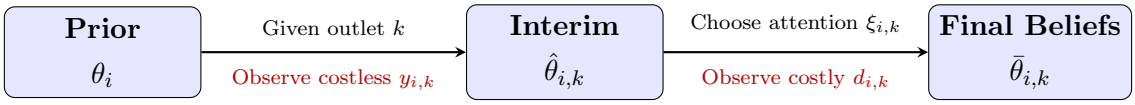
$$\overline{\theta}_k \sim \mathcal{N}\left(m_{2,k}(\xi_k), V_{2,k}(\xi_k)\right)$$

In our model, costly learning is important for voters because it helps them make more accurate decisions when deciding which party to vote for. Allocating further effort to distill the news and learn about one-self's policy stance is more valuable as interim uncertainty is larger and when political platforms are more different to each other.

The main difference here with respect to most of the literature is that how much information is available for a voter to refine her knowledge is endogenous and determined by the media outlets.

After acquiring both signals voters will then choose the policy q_j that is closer to his final beliefs. In the simplified version of the problem, we will assume that party platforms are exogenous and given by some preferred exogenous policy q_j^* for each party. For now we can keep the political stakes $\Delta u'(q)$ to be exogenous, but the outcome can be formalized through the lens of a rather standard probabilistic voting model (Persson and Tabellini, 2000).

For the sake of clarity, we include the summarized steps of the voter's learning problem below:



4.3 Traditional Media

The first type of outlet we have in the economy is Traditional Media. This type of outlet can be related to incumbent media sources such like TV, radio, newspapers, and even websites. We assume there is a monopolist traditional outlet, since competition between different these outlets is not the main focus of our paper. We can think of traditional media as a professional content producer that generates news/signals only about the common policy component μ. Traditional media would produce an imperfect signal about this component of the voters'

policy stance such that:

$$y_T = \mu + \nu_T, \qquad \nu_T \sim \mathcal{N}\left(0, \psi^{-1}\right). \tag{4.3}$$

We assume that Traditional Media's choice variable is the precision of the signal, $\psi \geq 0$. Thus, given a particular ψ, and using that the prior is $\mu \sim \mathcal{N}(m_0, \frac{1}{\tau_\mu})$ the interim variance for a traditional media outlet viewer is:

$$V_{1,T}(\psi) = \frac{1}{\tau_\mu + \psi} + \frac{1}{\tau_\epsilon} \tag{4.4}$$

Where τ_μ and τ_ϵ are the prior precisions on the common and idiosyncratic components, respectively.

Thus, a higher ψ corresponds to more precise professional reporting. It can be interpreted as greater investment in reporting, fact checking, investigation, editorial quality, or other activities that improve the accuracy of professional news production. Note this is telling us as well that, due to the informational constraints of the Traditional Media, direct viewers of this outlet cannot reduce any uncertainty about their idiosyncratic component.

We assume that producing a more precised report is costly. Particularly, assume the cost of a report is quadratic on signal precision:

$$C_T(\psi) = \frac{c}{2}\psi^2, \qquad c > 0.$$

Traditional Media does not choose households' attention directly. Instead, it anticipates that its choice of ψ affects the residual uncertainty faced by households and, therefore, their subsequent rationally inattentive information-acquisition decision.

To see how, let us return to expression (4.2) for the optimal attention allocation of a voter of media k. A viewer that directly consumed her information through traditional media (something that recall happens with probability $1-p$), will choose how much attention to allocate in order to reduce her interim uncertainty about the common component, which is a decreasing function of ψ.

Given the cap on the uncertainty that can be reduced about θ when listening to Traditional Media, given by $\bar{\xi}_T$, we have that optimal attention for those voters will be given by the following general expression:

$$\xi_T^* = \min\left\{\bar{\xi}_T, \max\left\{0, 1 - \frac{\chi_T}{V_{1,T}}\right\}\right\} \tag{4.5}$$

Where $\chi_T \equiv \frac{\kappa_T}{2(\Delta u)^2}$ depends positively on the marginal cost of attention κ_T and inversely

on the distance between the marginal utility evaluated in both political parties' platforms, $(\Delta u)^2$.

Expression (4.5) tells us that we could have different regimes of attention allocation in the second stage. depending on the relative magnitude of the interim variance and the informational ceiling given by the inability of Traditional Media to reduce information about the idiosyncratic component.

Note that whenever $\xi_T \geq V_{1,T}$ we will have $\xi_t^* = 0$, and no further learning would take place. This is related to the voter having little uncertainty left over to reduce through a second signal. Since $V_{1,T}$ is decreasing in quality, ψ, then a better quality reporting by Traditional Media would make it more likely that the voter does not need to further spend effort learning about θ.

On the other hand, when $\xi_T < V_{1,t}$ we will be in an interior solution, provided that optimal attention allocation does not surpasses $\overline{\xi}_T$. Once reaching that point, attention allocation becomes constant, with its value given by the binding constraint on learning from this media outlet. Whenever this is not the case, we have that as party becomes more polarized in their policies (greater political stakes) make voters willing to acquire more information, thus increasing attention. This relates to the fact that more polarized environments tend to correlate with higher levels of attention. Naturally, following our previous discussion, higher interim uncertainty will produce greater incentives to exert more costly attention.

Now turning to the Traditional Media's profits, we need to make a distinction between the types of voters that ultimately get content from the outlet. In our setting, we have that either voters go directly to traditional media for acquiring information, in which case these are a mass $(1-p)$ of *direct viewers*, or they consume their content from social media. In the latter case, we refer to these as *indirect users* of traditional media and the revenues it can get from these depend on the arrangement offered by social media.

Under platform entry, Traditional Media earns revenues both from direct viewers and from the use of its content within Social Media. We denote by $\lambda \in [0,1]$ the share of Social Media's revenues that are split with Traditional Media.[4]. This directly relates to the fact that the platform uses content produced (and owned) by the professional content creator (that is, Traditional Media), for which use it needs to compensate it for.

We take λ as an institutional parameter rather than solving for it endogenously[5]. It can

[4]Related work has studied how news aggregation affects publishers' traffic, profits, and incentives to produce quality content Dellarocas et al. (2013); Jeon and Nasr (2016). Our parameter λ is a reduced-form representation of how much of the value generated through intermediation is internalized by the original producer; it is not intended to reproduce the contracting environment of those papers.

[5]Modeling the determination of λ would require an additional bargaining or pricing problem. This is outside the scope of the benchmark, whose objective is to understand the consequences of media intermediation for information production.

summarize publisher-platform agreements, licensing arrangements, revenue-sharing rules, or regulation.

Although Traditional Media chooses the quality of its reporting, it might be the case that Social Media does not decide to fully display what this outlet is publishing. This is related to a *transmission quality* motive, in which the platform might display a mix of user generated content and Traditional Media produced content in its feed. We will be more precise about how this impacts voter learning when describing the Social Media's problem. For the moment being, it suffices with defining this weight as $\omega \in [0,1]$, where $\omega = 1$ means the Social media decides to display the whole precision of reporting chosen by Traditional Media.

Hence the revenues that Social Media gets, and potentially splits with Traditional Media, are described by a function of both attention and reporting quality, for a given ω. Let us call this function $R_S(\psi, \xi_S; \omega)$. We make for the moment no further assumptions on its particular form, but only that it is increasing on both its arguments.

Profits for traditional media can be expressed as the sum of the revenues generated by interactions from direct viewers and those coming from its content displayed in the Social Media platform, minus the costs of producing a precise reporting:

$$\Pi_T(\psi) = \underbrace{(1-p)R_T\left(\psi, \xi_T(\psi)\right)}_{\text{Direct users}} + \underbrace{p\lambda R_S(\psi, \xi_S(\psi); \omega)}_{\text{From Social Media}} - \quad C_T(\psi). \tag{4.6}$$

Where in the first term $R_T(\cdot)$ is a revenue functions that depends on information quality and the attention paid by direct viewers in the second stage. On the other hand, the transfers from the platform are a function of not only how traditional media content is displayed in Social Media's feed (ω), but also of the quality of that information, and how much attention is paid to acquire a further signal from Social Media. Furthermore, the share of the platform's revenues that are acquitted to Traditional Media scale this term. The figure below summarizes the Traditional Media's problem:

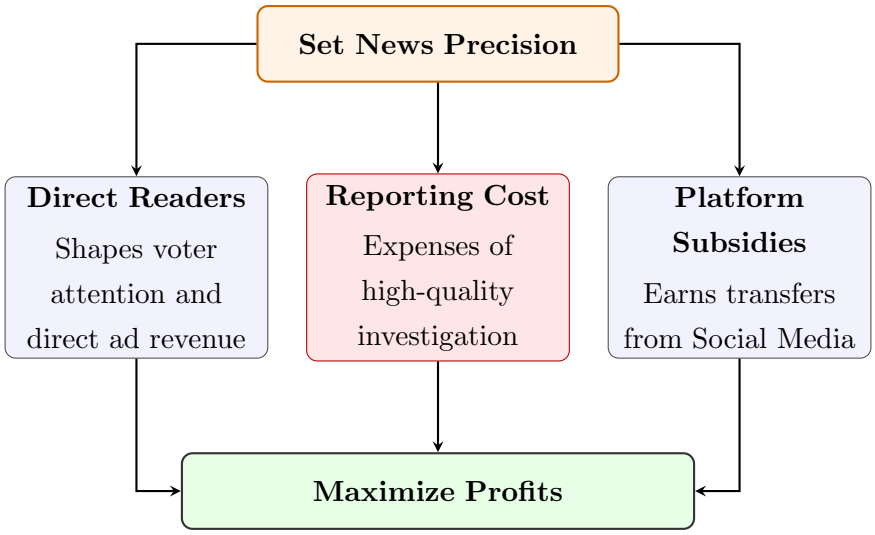


Figure 3: Traditional Media's Problem

4.4 Social Media

The Social Media entrant acts as a monopolist platform intermediating information between traditional media and viewers. As presented above, the Social Media platform receives content from both the traditional outlet and from users that join the platform (fraction p of the voters).

Users generate information about themselves through their activity on the platform. This information may come from content creation and sharing, but also from passive behavioral signals such as clicks, likes, follows, viewing time, and other interactions. As the platform's user base expands, the amount of information available to its recommendation technology increases, allowing it to infer user-specific interests more precisely. We capture this data externality in reduced form by assuming that the precision of the personalized signal is γp.

Hence, as more users join the platform, the easier is to produce back a more precise estimate of ϵ_i. Thus, the platform is able to send a social media viewer an estimate $\hat{\epsilon}_i$ of the form:

$$\hat{\epsilon}_i = \epsilon_i + \nu_S, \quad \nu_S \sim \mathcal{N}\left(0, \frac{1}{\gamma p}\right)$$

In other words, as more people join the social media platform, the variance is reduced further. The parameter γ measures how effectively the platform can extract personalized information from a given user base. The interaction γp captures the idea that a larger data environment can improve the precision of individualized information, in line with the broader literature on social data and data externalities Bergemann et al. (2022).

This is but only one side of the platform's signal. As we have mentioned, the other side

comes from the information already produced by the traditional media, which has original precision ψ. Professional information need not reach voters in exactly the same form when it is consumed through Social Media. Let $\omega \in (0, 1]$ to measure the effectiveness with which professional information is transmitted through the platform.

This implies that a Social Media user receives information on μ with effective precision $\omega\psi$. When $\omega = 1$, professional information is transmitted without loss, while for $\omega < 1$ the information reaching Social Media users is less precise than when the original report is consumed directly. The way in which this affects voter learning is by affecting the precision of the signal the Social Media viewers get about the common component.

Hence, we can think of Social Media as distributing a signal about the common component whose variance may be distorted away from what the Traditional media originally chose. Calling this signal $\hat{y}_T$, this feature amounts to the platform adding an additional noise η to the Traditional Media's reporting:

$$\hat{y}_T = y_T + \eta, \quad \eta \sim \mathcal{N}\left(0, \frac{1-\omega}{\omega\psi}\right)$$

Hence ω is a reduced-form transmission technology. It can capture differences associated with aggregation, headlines, summarization, ranking, or more fragmented exposure to professional news. This follows work on online news aggregation documenting that platform-mediated access changes how readers encounter and consume original news content (Chiou and Tucker, 2017; Athey et al., 2021).

Thus, the social media outlet is able to send a signal vector to its users about both components of θ_i. We can define this signals vector as $\mathbf{y}_{S,i}$ such that:

$$\mathbf{y_{S,i}} = (\hat{y}_T, \hat{\epsilon}_i) \tag{4.7}$$

The previous expression makes it clear that more precision from the traditional media outlet allows to acquire better information about the common component also through Social Media.

Both components of θ are conditionally independent. Thus, when updating their priors, voters would have a posterior that can be separated in both components:

$$V_{1,S}(\psi, p) = \frac{1}{\tau_\mu + \omega\psi} + \frac{1}{\tau_\epsilon + \gamma p} \tag{4.8}$$

Where the error terms are both normally distributed and independent such that:

$$\begin{pmatrix} \nu_T + \eta \\ \nu_S \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} (\omega\psi)^{-1} & 0 \\ 0 & (\gamma p)^{-1} \end{pmatrix} \right)$$

Given the nature of the Social Media's technology, we can write the optimal attention allocation in the costly state for a Social Media viewer as:

$$\xi_S^* = 1 - \frac{\chi_S}{V_{1,S}} \tag{4.9}$$

Here, χ_S is defined similarly as χ_T, and is thus a function of the marginal cost of attention and the distance between the exogenous political parties' platforms. Naturally, equation (4.9) characterizes the optimal attention allocated to a costly signal in an interior equilibrium.

Recall, following our discussions in the voter's problem that the cap on ξ_S is equal to $\bar{\xi}_S = 1$. This is due to the Social Media allowing the voter to reduce uncertainty about both of the components of θ_i. Thus, a sufficiently high ξ_S could allow the voter to reduce all the uncertainty about θ pertaining to the cognitive restrictions of voters. Naturally, there are other frictions that are outside the voters control, like the transmission quality of Traditional Media content in the platform's feed, ω, that blurs in a way what she actually learns from the original report through the costless signal.

Because of the assumed structure of the costly signal $d_{i,S}$, we are assuming that ω plays a role in distorting the reporting of Traditional Media for the costless initial signal voters receive. However, once the viewer decides to further exert costly attention to refine her knowledge of θ_i the chosen transmission ω does not affect her signal. That is, the transmission quality ω will have an impact on learning only through its effect on the interim uncertainty of voters.

Costly learning can be related to a process of further digesting and researching about the particular topic or news of interest. Likely, the voter can better inform herself by clicking in the actual source (i.e. the link to the Traditional Media's article that appeared in her feed) or spend more time looking at different articles or videos on the matter. Thus, she could reduce the whole uncertainty about θ, given there are no technological constraints whatsoever on what can be learnt from Social Media about both components of the policy stance.

One obvious question concerns the comparison we can establish between the attention allocations of both media outlets. For simplicity, let us assume that $\chi_S = \chi_T = \chi$. This implies that we assume that the marginal costs of attention are the same for both Traditional and Social media. Then, if we assume an interior solution for (4.5), we get the following

condition:

$$\xi_S^* \lessgtr \xi_T^* \quad \iff \quad V_{1,S} \lessgtr V_{1,T}$$

Thus, in general the relative optimal attention allocation across media outlets is not fixed. Rather, it depends on the relative uncertainties left after the first costless learning stage. The uncertainty left over after listening to social media's report decreases with ω. Thus, curating content in such a way that less of the original report of Traditional media is displayed will left higher uncertainty for Social media voters, thus increasing ξ_S, all else equal.

On the other hand, as more people use the Social media platform (an increase in p) or, alternatively, when Social Media has better personalization technology (higher γ), the remaining uncertainty for Social Media voters after the first signal decreases due to better personalization. This will make it more likely that Traditional Media voters are the ones exerting higher attention effort in the additional signal.

One clean difference on attention allocation arises through evaluating the impact that more dispersion in political parties' preferred policies (i.e. a lower χ) has on the incentives to allocate attention. This leads to the following proposition:

Proposition 1. *For sufficiently high political stakes, Social Media voters will always allocate more costly attention than Traditional Media's voters.*

The proof for this entails looking at (4.5) and (4.9) and taking the fact that as political stakes grow larger then $\chi \to 0$. Thus, comparing both conditions we will have:

$$\lim_{\chi \to 0} \xi_T^* = \frac{a(\psi)}{a(\psi) + b} \qquad \lim_{\chi \to 0} \xi_S^* = 1$$

One direct implication of this, is that whenever party ideologies are sufficiently far from each other, we could expect a rise in attention overall, while the relative costly learning tilts more towards Social Media users.

One further clarification is that whenever Traditional Media voters hit their learning constraint the ordering of attention allocation across outlets is not direct. To see this, assume a situation in which $\xi_T = \bar{\xi}_T$. In that case, Social Media voters will exert **more** costly attention whenever the following holds:

$$\chi \leq \frac{bV_{1,S}}{V_{1,T}}$$

This inequality might not hold if uncertainty about the idiosyncratic component is too low (lower b), or whenever interim uncertainty for Social Media voters is relatively small. The

latter is more likely to happen if ω is low. That means, whenever too much of the original precision of the Traditional media is distorted away by Social Media's transmission quality.

Given these differences in learning, we can describe the problem of Social Media. This intermediary will have revenues R_S that, as stated when describing the Traditional Media's problem, depend positively on the attention allocation ξ_S, on Traditional Media's transmission quality ω and news precision ψ. These revenues are scaled by the share of Social Media voters, p, and we assume it retains the fraction $1-\lambda$ that are not transferred to Traditional Media to compensate it for its content in the platform.

A general form of the profit function needs to include a cost of curating content in favor of Traditional Media. We approach this by a function $K(\omega)$ that is increasing and convex in ω, and that satisfies $K(0) = 0$, such as the following $K(\omega) = \frac{\zeta}{2}\omega^2$, where $\zeta > 0$ is a cost-scaling parameter. This cost can be interpreted broadly. Increasing ω may require greater curation, give more prominence to original professional content, or reduce the platform's ability to prioritize other forms of content. For instance, high-fidelity retransmission of Traditional media's content may require richer licensing agreements, content formatting, fact-preserving summaries, editorial integration, or technological resources. Also, one can think that a higher ω has also some cost of opportunity for the platform, since we can interpret a lower ω to be related to given relatively more room for user-generated content in the feed. A higher predominance of Traditional Media displaces this type of content, which we do not explicit model in the profits.

Thus, the objective function of Social Media can be written as follows:

$$\Pi_S = p(1-\lambda)R_S(\psi, \xi_S; \omega) - K(\omega) \tag{4.10}$$

We assume the platform can choose its content visibility or transmission quality ω, while the revenue share λ will stay fixed for the benchmark. As described above, λ summarizes the institutional relationship between Social Media and the original content producer. Endogenizing this parameter would require introducing an additional stage of bargaining between both media outlets. We instead take the arrangement as given and ask how the extent to which Traditional Media internalizes the value of its content affects information production.

The main elements of the Social Media's problem are summarized below in figure (4)

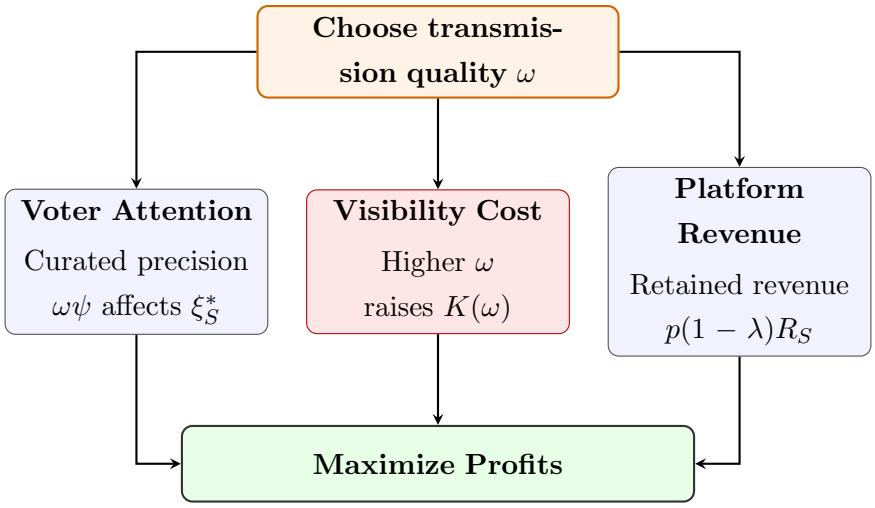


Figure 4: Social Media's Problem

5 A Traditional Media's Monopoly Benchmark

To start with the exposition of the main dynamics that result from the previously described interactions it is useful to begin with an economy in which Traditional Media is the only source of information to voters. This is related to a pre-Social Media economy, and isolates the main channels through which Traditional media will choose its precision absent competition, or in this case intermediation, by another information source.

Under monopoly we have $p = 0$, so that all voters get their information from Traditional Media. The traditional media's profits are equal to the general expression (A.1), but without the second term, which accounts for the revenues coming from the content broadcasted through social media. Thus, in this case the objective of traditional media is just their direct revenues minus its precision cost:

$$\Pi_T^M(\psi) = R_T\left(\psi, \xi_T(\psi)\right) - C_T(\psi).$$

It is time to give the revenue function a particular form that allows to conduct some numerical exercises. We think a particularly simple reduced-form revenue function is one that is affine in both arguments, such as:

$$R_T\left(\psi, \xi_T\right) = r\psi + A\xi_T,$$

The parameter $r > 0$ captures the reduced-form commercial or reputational value of professional reporting quality, while A captures the commercial value of voter attention. We do not

model the underlying advertising market explicitly. Evidence and theory in the media literature provide several reasons why information quality and audience reach can affect media revenues (Gentzkow and Shapiro, 2006; Petrova, 2011). Also, for revenues being positively affected by attention from users Jeon and Nasr (2016).

5.1 Discussion on the revenue function

Economically, this form means that, from the point of view of the traditional media, precision and reader engagement are substitutes. Likely, a more general form would allow for some imperfect substitutability (like in a CES function); however, we think a parsimonious example like this allows to have a better characterization fo the main effects in the model. We think it is useful to think about this as a case in which good investigative reporting is a good sign of trustworthiness, but a *too lenghty* reporting is not desirable for further engagement.

This form is also grounded in the literature on attention allocation, both from a theory and an empirical perspective. Based on Matysková and Montes (2023) we can think of a media outlet as being the sender deciding an information structure through its reporting precision. In their paper, a receiver can exert costly attention effort to acquire a second signal to refine her knowledge about a payoff-relevant state. Although their paper involves a Bayesian persuasion framework grounded in the seminal work by Gentzkow and Kamenica (2011), their work speaks to a fundamental tension that might exist between the reporting precision and subsequent allocation of attention by a receiver.

Other papers, like Galperti and Trevino (2020) discuss possible ways in which information precision and attention devoted to a source need not go in the same direction, although in their framework, these forces are dictated by how much the receivers care about coordinating their actions to the others'. Also importantly, is that their model assumes a revenue function for media that depends positively on the attention allocated to a source (a media outlet in our language). They discuss how this is related to the fact that greater attention (higher ξ in our paper) would mean more ads can be showed to viewers, and thus more ad revenues can be extracted from advertisers.

Lastly, Serra-Garcia (2026) uses experimental evidence to support the idea that there exists a quality-attention tradeoff in how information is presented at first to users and the subsequent engagements or interactions with the news they were presented with.

In a related note, her paper also provides evidence on how the monetary incentives to produce *more viral* content has a negative impact on the quality of the summaries presented to readers. This last bit becomes particularly relevant when we introduce competition and

interactions between the platform and the traditional media. However, in the monopolist case we can immediately see as a reduction in the parameter r.

5.2 Solving the Monopolist Problem

The monopoly problem is therefore

$$\max_{\psi \geq 0} \Pi_T^M(\psi) = r\psi + A\xi_T(\psi) - \frac{c}{2}\psi^2.$$

The solution to its problem is described by the following first-order condition, that makes transparent the trade-offs the outlet faces when choosing how much precision to give to its report.

$$r + A \underbrace{\frac{d\xi_T^*}{d\psi}}_{<0} - c\psi = 0. \tag{5.1}$$

Thus, we can describe the optimal quality of reporting as ψ^*, which represents the solution to (5.1). The actual form of the solution depends on the attention allocation regime we are at. Particularly, if we are at an attention ceiling or not.[6]

Lemma 1. *For a given political environment, an increase in professional reporting precision weakly reduces the optimal costly attention of Traditional Media viewers* $\frac{d\xi_T^*}{d\psi} < 0$.

Moreover, it could well be that optimal attention allocation is zero, as showed in (4.5). In this latter case we have that the optimal quality of reporting is independent of the further attention allocated and equal to $\frac{r}{c}$.

For a positive attention allocation, the actual closed form solution for ψ^* is harder to get. Note that attention is at its ceiling whenever $\chi_T \leq b$. This condition is independent of ψ. This implies that, fixing a marginal cost of attention, once political stakes become sufficiently high (i.e. political parties become more differentiated), the optimal attention allocated to further signals will be constant for any reporting precision.

Let us assume that the attention constraint is non-binding, such that $\xi_T > b$. For a situation like this the optimality condition becomes:

$$r - c\psi - \frac{A\chi_T}{[1 + b(\tau_\mu + \psi)]^2} = 0 \tag{5.2}$$

This last expression illustrates the main trade-offs that the Traditional Media faces when

[6]Formally, a global optimum needs to lie in a region such that $0 \leq \psi^* \leq \frac{r}{c}$. For the Traditional Media reporting precision above $\frac{r}{c}$ cannot be optimal because direct reporting revenues net of costs decline, while costly attention does not increase with reporting precision.

picking its optimal reporting quality. Increasing reporting quality brings extra revenues equal to r, while increasing its costs. However, the further increase in precision will decrease profits through lower attention allocated to further information reported by the outlet. This last part corresponds to the third term of equation (5.2).

Alternatively, when $\chi_T \leq b$ the attention constraint binds and the optimality condition becomes:

$$r - c\psi - \frac{A\tau_\epsilon}{[\tau_\epsilon + \tau_\mu + \psi]^2} = 0 \tag{5.3}$$

Which, by our previous discussion, makes it transparent that for this attention regime, further changes in political stakes or attention marginal cost do not alter the monopoly profits.

The distinction between the learning regimes are clearer in figure 5 below. The figure on the left shows the optimal ξ_T, which is the fraction of interim uncertainty reduced after acquiring the costly signal. On the other hand, the figure on the right panel plots the final variance of θ_i for the voters. We use the fact that political stakes, appropriately scaled, are the inverse of χ

We see that for the given calibration there is a range of sufficiently low political stakes that induce zero costly learning by the voters. After some threshold, we enter into a region in which the optimal attention is increasing in political stakes. Here, since parties are becoming more differentiated the voters have more incentives to learn about their true θ so as to choose the party that aligns closer to their preferences. Finally, political stakes rise to a point in which voters hit their learning constraint. They would have liked to exert more costly attention, as showed by the dotted line, but since they cannot learn about ϵ_i, the reduction in uncertainty is capped at a flat value given by $\bar{\xi}_T$.

The figure on the right panel of figure 5 shows us the counterpart of ξ. Given that we do not exert any costly attention for small political stakes there is no further reduction in the variance. Thus, the final variance of θ equals the interim variance for voters. In the interior learning region, we see a smooth decline in final variance, as voter's have higher ξ_T. Finally, their constraint binds. No further reduction in uncertainty is possible and final variance equals σ_ϵ^2, which has been normalized in this example to 1.

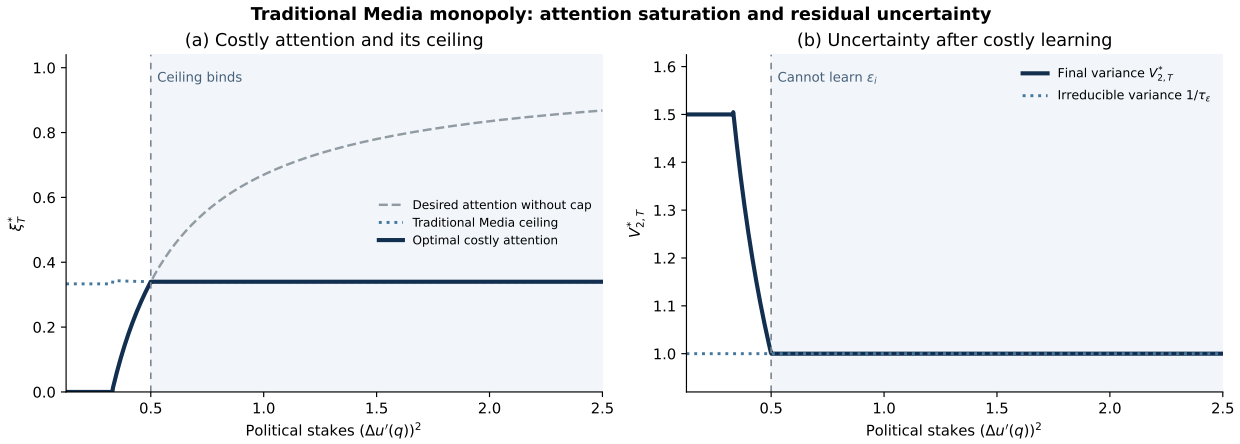


Figure 5: Left: desired attention without the technological restriction, feasible ceiling, and actual attention. Right: final uncertainty about the voter's preferred policy. Reporting precision is reoptimized at each level of political stakes.

Naturally, depending on the parametrization we choose we can have situations in which voters never hit their attention ceiling. While we can have situations in which they always do and the optimal precision never depends on political stakes. Given that we can write the final variance after both signals have been received as $V_{2,T} = V_{1,T}(1 - \xi^*_T)$ then:

$$V_{2,T} = \begin{cases} V_{1,T} & \text{if } \chi_T \geq V_{1,T} \\ \chi_T & \text{if } \ b < \chi_T \leq V_{1,T} \\ \text{b} & \text{if } \chi_T \leq b \end{cases}$$

These changes also affect Traditional Media's reporting incentives. As illustrated in Figure (6), the monopoly initially chooses reporting precision $\frac{r}{c}$, since voters acquire no further information. When costly learning becomes attractive, the monopoly has an incentive to reduce its reporting precision to encourage additional attention. Within the interior-learning regime, reporting precision subsequently increases with political stakes, as the negative effect of reporting quality on costly attention becomes weaker.

Finally, once voters reach their informational ceiling, political stakes no longer affect the monopoly's profits or optimal reporting precision. This establishes a useful benchmark: even when voters have strong incentives to learn, the informational restrictions of Traditional Media prevent them from eliminating uncertainty about the idiosyncratic component of their preferred policy position.

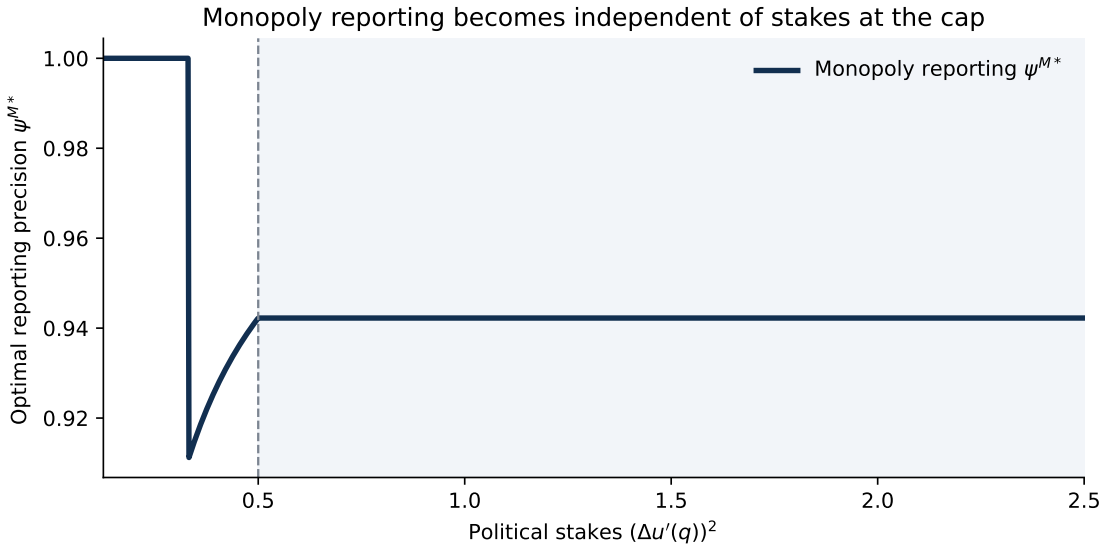


Figure 6: Optimal reporting as a function of political stakes.

6 Equilibrium with Social Media

For the main section of our paper, we study the effect that Social Media entry has on the monopolist's benchmark. In this framework, a fraction $p \in (0, 1)$ of the population obtains news through the platform, while the remaining fraction $1 - p$ consumes Traditional Media directly.

Traditional Media continues to choose only ψ. The platform chooses the transmission quality ω it places on Traditional Media content content. This will shape how large is the compensation to Traditional Media for participating in the Social Media's feed. To fix ideas, we will explore first a setting in which Social Media does not make any strategic decision. That is, one in which ω is taken as given. This will allow us to explore the effects that introducing these objects have on the Traditional Media's choices.

All the way through our analysis we will assume a linear form for the Social Media's revenue function. Given we have been focusing on how the platform's transfers to Traditional media molds its precision choices, we make this function to only depend on the professional content generated by Traditional Media. A richer problem could include explicitly the content generated by other agents such as influencers or any sort of user-generated content. These considerations remain out of the analysis.

Thus, the ad revenue function for Social Media follows a similar shape as that of Tradi-

tional Media to make the comparison as cleanly as possible:

$$R_S(\psi, \xi_S) = r\omega\psi + A\xi_S.$$

The term $r\omega\psi$ reflects the effective quality of professional information reaching Social Media users, this can be seen as reputation for the transmission quality investment from S. While $A\xi_S$ captures the commercial value associated with user attention or engagement. Note that we are deliberately making both r and A to be equal across outlets. These revenue functions we are using are deliberately reduced form. Both r and A are not advertising prices chosen by the outlets. They summarize in either case the commercial value generated by information quality and user engagement.

6.1 Exogenous Transmission Quality

Once ω is fixed at some value, we have that the problem of Traditional Media amounts to choose ψ to maximize (A.1). The optimality condition is in the case of an interior optimum given by the first order condition.

Proposition 2 (Interior optimal ψ^*)**.** *The interior solution for the T outlet is defined by the following first order condition*

$$(1-p)\left[r + A\frac{\partial \xi_T^*}{\partial \psi}\right] + p\lambda\left[r\omega + A\frac{\partial \xi_S^*}{\partial \psi}\right] = c\psi. \tag{6.1}$$

The first term on the LHS of (6.1) is identical to what we had on the Traditional-Media-only case in expression (5.1), just scaled by the share of Traditional Media viewers $(1-p)$. If we ignore the share of viewers, the only change with respect to the monopoly case is the term $\lambda\left[r\omega + A\frac{\partial \xi_S^*}{\partial \psi}\right]$. The second bracketed term is the corresponding marginal return generated by professional information consumed through Social Media, of which Traditional Media captures only the fraction λ.

This term tells us that, for given λ and ω, an increase in reporting quality increases revenues in a magnitude $\lambda r\omega$, at the cost of reducing attention (and thus revenues) that equals $\frac{\partial \xi_S^*}{\partial \psi} < 0$

To simplify the exposition, define $M_T \equiv r + A\frac{\partial \xi_T^*}{\partial \psi}$ as the marginal commercial return generated by an additional unit of professional precision through a direct reader, while $M_S \equiv r\omega + A\frac{\partial \xi_S^*}{\partial \psi}$ is the corresponding return generated through a Social Media user. Thus we can describe the optimal reporting quality as the solution of a fixed point problem such that:

$$F(\psi^*, p) = (1-p)M_T + p\lambda M_S - c\psi^* = 0 \tag{6.2}$$

Assuming that the objective is strictly concave in the region of interest, we will have that a sufficient condition for higher Social Media penetration (a higher p) to reduce optimal reporting quality is that $M_T > \lambda M_S + p\lambda \frac{\partial M_S}{\partial p}$. This requires a direct Traditional Media user to generate a larger marginal return from better professional reporting than the revenue-share-adjusted marginal return generated by a Social Media user.

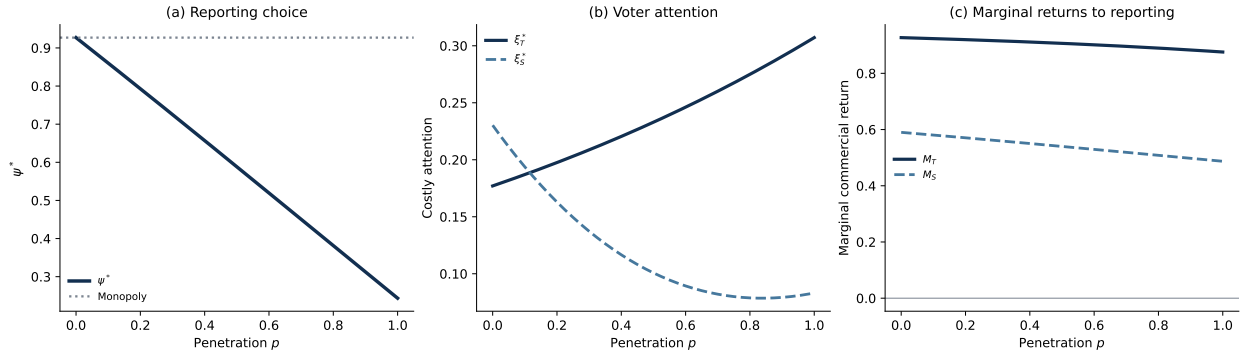


Figure 7: Effects of Social Media penetration on Traditional Media reporting, the two groups' attention, and marginal commercial returns M_T and M_S. Baseline parameters: $\tau_\mu = 1$, $\tau_\epsilon = 1$, $r = 1$, $A = 0.5$, $c = 1$, $\kappa_T = 1$, $\kappa_S = 1$, $\gamma = 1$, $\lambda = 0.5$, $\omega = 0.65$, $[\Delta u\prime(q)]^2 = 0.4$. Only penetration p varies; ω is exogenous and λ is fixed.

Thus, under these conditions, Social Media penetration shifts audience weight away from the direct channel and toward the less profitable intermediated channel, weakening Traditional Media's incentive to invest in professional information. This mechanism is related to the literature showing that aggregation can change the incentives of original content producers to invest in quality (Dellarocas et al., 2013; Jeon and Nasr, 2016).

A particularly clean comparative static comes from the fact that greater Social Media penetration, keeping everything fixed will result on lower levels of reporting quality when political stakes increase. This interaction is better seen through the lens of figure 8. There, the optimal reporting precision increases for some range of political stakes. After this, there is a point in which Traditional Media voters hit their learning ceiling (remember figure 5) and precision stays constant for every level of political stakes. Note, however, that chosen precision is always smaller for higher values of p. This points at the fact that as more people get their news from Social Media, the quality of reporting by Traditional Media decreases due to less reliance on their direct viewers.

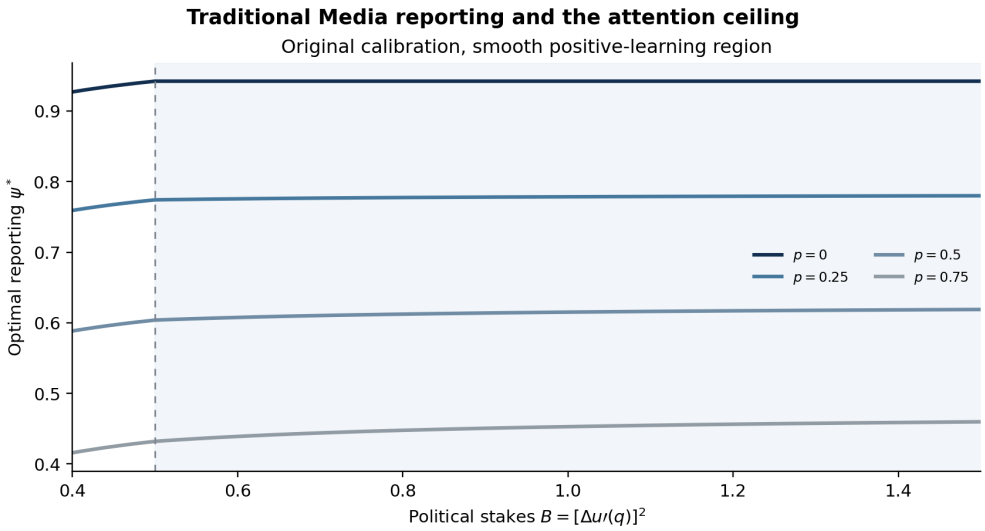


Figure 8: Effect of changes in political stakes on optimal reporting quality for different levels of Social Media penetration. Baseline parameters: $\tau_\mu = 1$, $\tau_\epsilon = 1$, $r = 1$, $A = 0.5$, $c = 1$, $\kappa_T = 1$, $\kappa_S = 1$, $\gamma = 1$, $\lambda = 0.5$, $\omega = 0.65$.

In figure 7 we see how greater share of Social Media users monotonically decreases Traditional Media's reporting quality. One interesting aspect is that, as the middle panel shows, it could be the case that for low levels of p the users of Social Media exert higher attention than the ones from Traditional Media, given the lower precision of the idiosyncratic signal.

This last part can be seen in more detail once we plot the chosen precision against different values of γ below in figure 9. We can immediately see that the chosen ψ is not continuous for any value of γ. This is due to a change in learning regimes that result whenever the personalization technology of Social Media becomes better. Note that a higher γ means that for every person that joins the platform, the Social Media outlet is able to report an even more precise signal about ϵ_i. This decreases overall uncertainty, and thus the incentives to allocate costly attention to further signals. The jumps we see, for a given level of p, are due to Social Media voters entering into a no-costly-learning regime. In those cases, their optimal ξ_S equals zero, which represents a discrete jump in the Traditional Media's objective. It is clear from expression (6.1) that when $\xi_S^* = 0$, the marginal return on ψ increases, since the negative derivative term disappears.

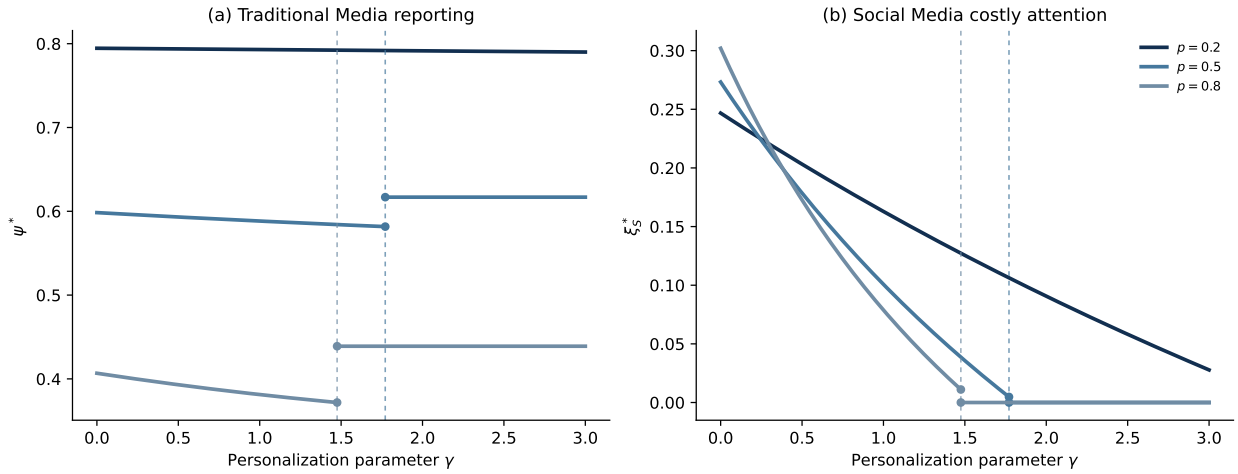


Figure 9: Personalization and the Traditional Media reporting choice, with exogenous transmission. Baseline parameters: $\tau_\mu = 1$, $\tau_\epsilon = 1$, $r = 1$, $A = 0.5$, $c = 1$, $\kappa_T = \kappa_S = 1$, $p = 0.5$, $\lambda = 0.5$, $\omega = 0.65$, $[\Delta u\prime(q)]^2 = 0.4$. We vary γ for $p = 0.2$, $p = 0.5$, $p = 0.8$. Dashed vertical lines show switches of the global optimum from positive to zero Social Media costly attention.

On a second note, the effect of higher λ on precision is more straightforward. Provided that $M_S > 0$ we will have that greater revenue sharing increases information quality, such that $\frac{d\psi^*}{d\lambda} > 0$. Thus, greater revenue sharing increases information production whenever Traditional Media content has positive marginal commercial value on Social Media.

This relationship is illustrated in Figure 10. Moreover, we can see this clear positive relation is preserved as we vary the level of platform penetration. Provided λ stays as a predetermined value, a larger share of platform viewers will decrease the optimal reporting quality of Traditional Media for any possible value of the Social Media revenues the Traditional Media gets.

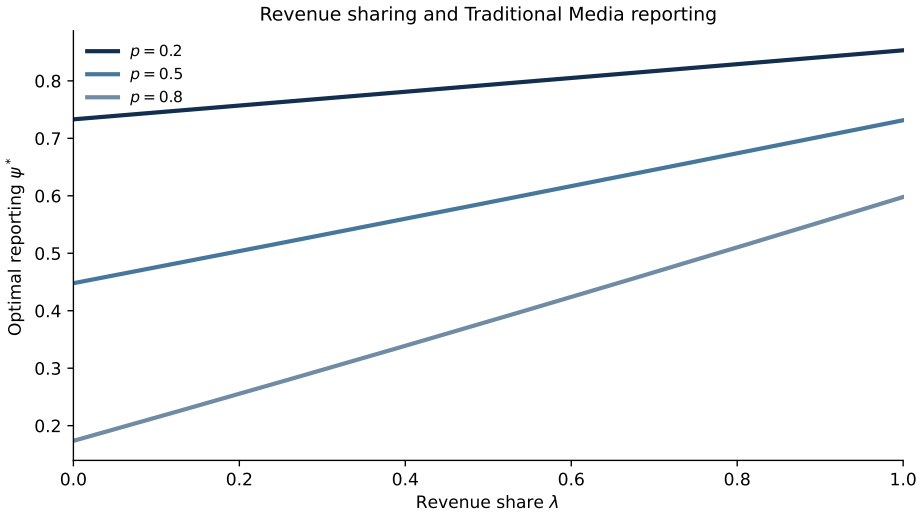


Figure 10: Optimal Traditional Media reporting and the revenue share λ. Parameter values: $\tau_\mu = 1$, $\tau_\epsilon = 1$, $r = 1$, $A = 0.5$, $c = 1$, $\kappa_T = 1$, $\kappa_S = 1$, $\gamma = 1$, $[\Delta u\prime(q)]^2 = 0.4$, $\omega = 0.65$. The lines vary the platform penetration rate ($p = 0.2$, $p = 0.5$, $p = 0.8$); the horizontal axis varies λ, holding all other parameters fixed.

Lastly, the effects that changes in ω have on reporting quality of Traditional Media are a bit more nuanced. Clearly, a higher ω rises the incentives to provide higher precision, given that it increases Traditional Media's revenues by $pr\lambda$. However, we have seen that there exists a trade-off between initial precision of the costless signal and subsequent engagement (costly learning). An increase in ω also depresses profits through a lower future ξ_S. Thus, the final result may depend on the chosen parameter region.

Particularly, for our example the relation between ω and ψ is always positive. Moreover, once we explore the interaction between λ and ω, we find that as ad revenue shares transferred to Traditional Media fall (lower ψ), chosen reporting quality by Traditional Media might still increase in ω.

Particularly, for high values of ω it could happen that Social Media users reduced initial uncertainty in such a magnitude that they find it optimal to not acquire any further costly signal. For such a combination of ω and λ the derivative of ξ_S^* with respect to ψ disappears from (6.1) and the incentives to increase ψ jump discretely. These can be see in the lines corresponding to higher values of λ in figure (11).

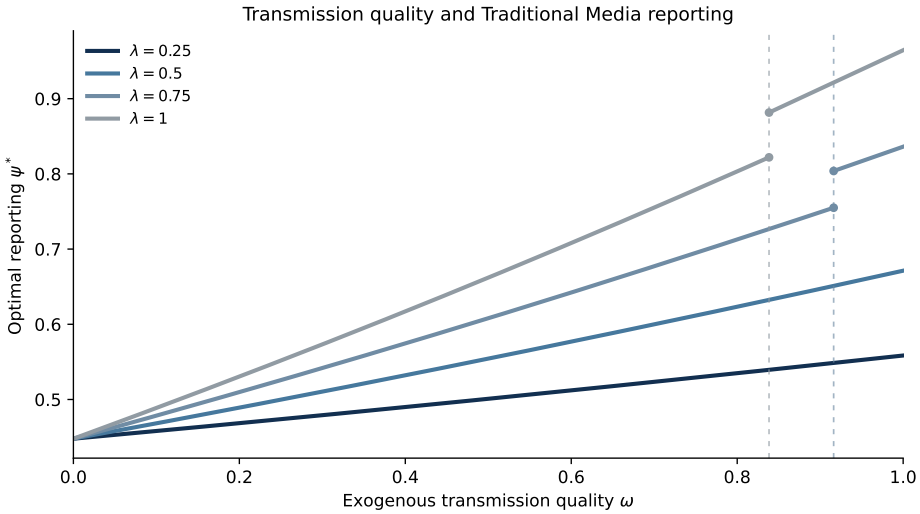


Figure 11: Optimal Traditional Media reporting and exogenous transmission quality ω. Parameter values: $\tau_\mu = 1$, $\tau_\epsilon = 1$, $r = 1$, $A = 0.5$, $c = 1$, $\kappa_T = 1$, $\kappa_S = 1$, $\gamma = 1$, $[\Delta u\prime(q)]^2 = 0.4$, $p = 0.5$. The lines vary the revenue share ($\lambda = 0.25$, $\lambda = 0.5$, $\lambda = 0.75$, $\lambda = 1$); the horizontal axis varies ω, holding all other parameters fixed. Dashed lines mark changes in the global optimum between positive and zero Social Media costly attention.

We must read the previous partial results as, at most, partial equilibrium effects of different variables introduced by Social Media's presence on the chosen precision quality of the Traditional Media. We have seen there are mechanisms showing that Traditional Media reports less precise news whenever the share of Social Media users increase. Also, party polarization, in the form of political stakes tends to have an effect of increasing both reporting quality of Traditional Media and the demand for further information. The last effect is related to the fact that voters perceived that as parties become more different, voting for the "wrong" alternative becomes costlier. They tend to seek to reduce more uncertainty to reduce the probability of these mistakes.

However, since Traditional Media's technology caps voter learning to being unable to learn about their idiosyncratic component, we may have that more polarized environments (those with higher political stakes) lead to greater demand for information but may also lead to more dispersion of beliefs on the voters' side, something that can be associated to higher voter polarization in beliefs.

Traditional media profits will vary depending on p. Now S media increases profits that T gets, but decreases also the engagement $\xi_{i,S}$ for the second signal by increasing by γp the precision of the fist signal. Then the main comparison is as follows.

Proposition 3 (Traditional Media's profits under Social Media entry)**.** *Let ψ^M denote Tra-*

ditional Media's optimal reporting precision under monopoly. Social Media entry weakly increases Traditional Media's maximized profits if

$$\lambda R_S(\psi^M, \xi_S^*; \omega) \geq R_T(\psi^M, \xi_T^*), \tag{6.3}$$

where ω is fixed under exogenous transmission.

The condition is sufficient but not necessary. In equilibrium, Traditional Media generally chooses a different reporting precision after Social Media entry. In particular, lower reporting precision reduces production costs and can partially compensate for the loss of revenues from direct viewers. Therefore, Traditional Media may benefit from S entry even when the revenue condition in the proposition is not satisfied by simply adjusting ψ.

In the next section we will explore the full equilibrium of the game and see the consequences for the dispersion of voters' final beliefs, at which point we will discuss what we mean for polarization in the context of this framework.

6.2 Social Media's Transmission Decision

In this section we solve for the full equilibrium of the game, focusing on the actions of the media outlets and voter learning. We have some given political stakes that affect costly learning through expressions (4.5) and (4.9). The form of these expressions already take into account that voters want to learn to make a better informed decision about which party to choose. Thus, in this part we decide to focus on the novel part of the paper. Namely, how the endogenous choices of two heterogeneous media outlets shape voter learning and the subsequent societal beliefs under which elections are held.

To recap, recall the game unfolds sequentially as follows:

1. Traditional Media chooses the precision ψ of professional information.
2. Social Media observes ψ and chooses transmission quality ω.
3. Voters receive costless information from their respective outlets.
4. Social Media users additionally receive personalized information.
5. Voters choose costly additional information acquisition.
6. Commercial revenues are realized.

Thus, Social Media when choosing optimally ω does so **after** Traditional media has already picked its desired reporting precision ψ. The sequential structure reflects the idea

that information must first be produced before Social Media can determine how effectively that content is transmitted to its users. The solution concept is subgame-perfect equilibrium, with Bayesian updating by voters, and the game is solved by backwards induction.

Before a thorough description of the equilibrium, let us state the problem of the Social media as having to choose $\omega \in [0,1]$ to maximize its profits. Formally, Social media chooses how much of the Traditional Media's precision to preserve in its feed. Calling this optimal choice ω^* then:

$$\omega^*(\psi) \in \arg\max_{\omega\in[0,1]} \left\{ p(1-\lambda)\left[r\omega\psi + A\xi_S^*(\psi,\omega,p)\right] - \frac{\zeta}{2}\omega^2 \right\}. \tag{6.4}$$

Then, we have that a subgame-perfect equilibrium is a pair (ψ^*, ω^*) such that:

1. for every ψ, $\omega^*(\psi)$ solves Social Media's problem in expression (A.2)
2. Traditional Media chooses ψ^* to maximize its profits from (A.1), anticipating $\omega^*(\psi)$;
3. Equilibrium transmission quality is $\omega^* = \omega^*(\psi^*)$.

Whenever we have an interior solution, the optimal transmission quality of Social Media will be characterized by the following first order condition.

Proposition 4. *For an interior solution, the first-order condition for social media outlet is*

$$p(1-\lambda)\left[r\psi + A\frac{\partial \xi_S^*}{\partial \omega}\right] = \zeta\omega. \tag{6.5}$$

The first term inside the brackets captures the direct benefit from transmitting more of the professional information, $r\psi > 0$, while $A\frac{\partial \xi_S^*}{\partial \omega} < 0$ captures the loss of subsequent user attention generated by better costless information.

Thus Social Media trades off the increased revenues from transmitting with better fidelity Traditional Media's information, against the associated reduction in subsequent attention and its cost of increasing transmission quality.

Our condition (6.5) defines implicitly the optimal transmission quality of Traditional Media's report by Social Media. This will be described by $\omega^* = \omega^*(\psi)$. Hence, we make explicit the feedback it exists between Traditional Media's chose precision and how the Social Media decides to display its report in the feed[7].

The equilibrium effect that better precision by Traditional Media has on Social Media's transmission fo this information is not unambiguous. As long as the additional revenues

[7]We have that under the condition $p(1-\lambda)A\frac{\partial^2 \xi_S}{\partial \omega^2} - \zeta < 0$ the objective function is strictly concave and we can guarantee that, for an interior solution, the maximum is unique.

coming from greater precision are positive, ω will be higher. Note that this is no longer the case whenever the additional rises in precision driven by higher ω reduce further costly attention sufficiently enough[8]

6.3 Traditional Media's Optimal Response

. Given that now the choice of ω is a strategic decision by Social Media, the traditional outlet now incorporates in its optimal reporting quality choice an additional term. Particularly, the first order condition of the Traditional Media can be written now as follows.

$$\underbrace{(1-p)\left[r + A\xi^*_{T,\psi}\right] + p\lambda\left[r\omega^* + A\xi^*_{S,\psi}\right]}_{(*)} + \underbrace{p\lambda\left[r\psi + A\xi^*_{S,\omega}\right]\frac{d\omega^*}{d\psi}}_{(**)} = c\psi \tag{6.6}$$

The term $(*)$ is exactly the same the one we had in (6.1) for an exogenous ω. What is added here is the extra term $(**)$. This will account for the general equilibrium effects of a change in optimal precision by Traditional Media. It captures the fact that Traditional Media understands that changing the quality of the information it produces may induce Social Media to change the way that information is transmitted in its feed.

Using (6.5) we see that $(**)$ becomes $\frac{\lambda\zeta\omega^*}{1-\lambda}\frac{d\omega^*}{d\psi}$. Therefore, when $d\omega^*/d\psi > 0$, the endogenous transmission response strengthens Traditional Media's incentive to produce precise information.

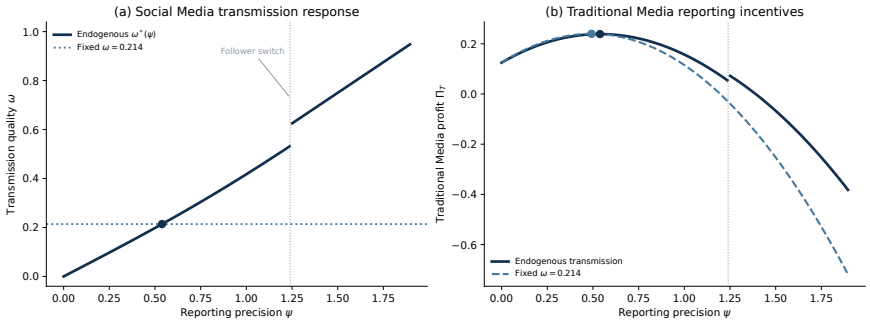


Figure 12: Strategic transmission and reporting incentives. Panel (a) shows Social Media's optimal transmission choice after observing the precision of Traditional Media reporting. Panel (b) compares Traditional Media's profits when it anticipates this response with profits when transmission is exogenously fixed. Parameters: $\tau_\mu = \tau_\epsilon = r = c = \gamma = \kappa_T = \kappa_S = 1$, $A = 0.5$, $p = \lambda = 0.5$, $\zeta = 0.5$, and $[\Delta u'(q)]^2 = 0.4$.

[8] A simple application of the Implicit Function Theorem formalizes this. Note that at the optimum $\frac{d\omega^*}{d\psi} = -\frac{p(1-\lambda)\left[r+A\xi^*_{S,\omega\psi}\right]}{p(1-\lambda)A\xi^*_{S,\omega\omega}-\zeta}$. This derivative is positive whenever $r + A\xi^*_{S,\omega\psi}$ is positive, given the denominator is negative under concavity of ξ.

Better precision of reporting by Traditional Media, for the region of parameters satisfying our condition above, will increase the transmission quality chosen by Social Media. The intuition is that the platform is willing to reward more a better quality reporting, all else equal. For Traditional Media this decision must balance the increased revenues from this better reporting with the costs of producing it. Panel (b) in figure 12 makes it clear that for our case profits are, for a wide range of ψ strictly concave in precision. The equilibrium values are showed with the points over the curves. Hence, we conclude that generally better reporting quality of Traditional Media tends to increase the optimal ω by Traditional Media.

A related aspect is how the revenue share parameter λ changes both outlets' responses. Figure 13 illustrates this. The relation between optimal reporting and λ is concave, with maximum at $\lambda = 0.5$. Interestingly, as the transfers to Traditional Media are higher, the reporting quality tends to decrease, something that contrasts with figure 10. This is due to Social Media not wanting to induce more precision of reporting, since it will further decrease ξ_S. Thus, it optimally reduces ω, as shown in panel (b). This impacts the revenues of greater precision of reporting for Traditional Media in such a way that the marginal value of better reporting decreases. Hence, the feedback between Traditional Media's precision and optimal transmission fo this information by Social Media creates a non-linearity of revenue shares on quality of reporting.

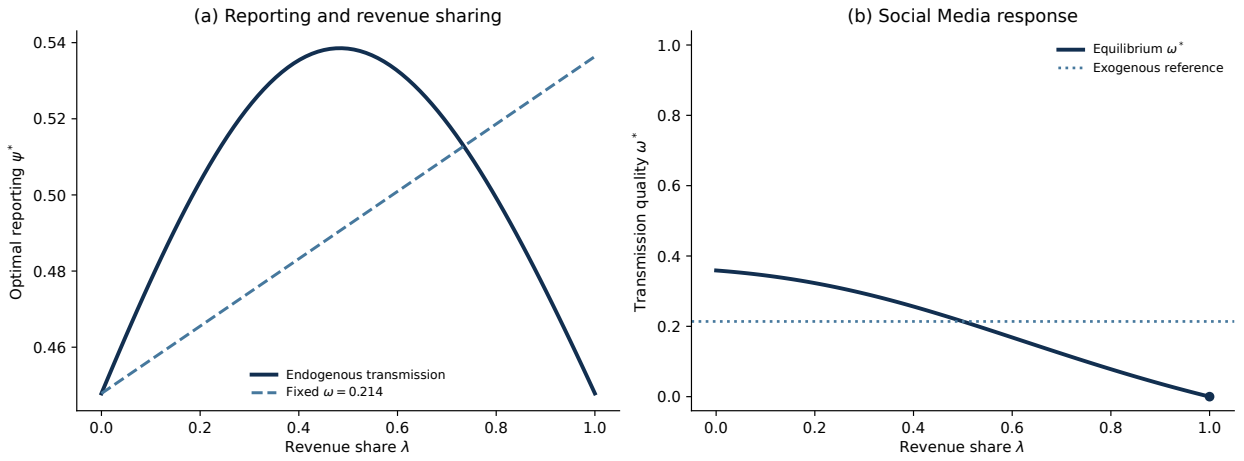


Figure 13: Revenue sharing with exogenous and endogenous transmission quality. Panel (a) plots globally optimal Traditional Media reporting as λ varies; panel (b) plots Social Media's equilibrium transmission choice. In the fixed-transmission benchmark, ω is held at the baseline endogenous equilibrium value, approximately 0.214, for all λ. The other parameters are $\tau_\mu = \tau_\epsilon = r = c = \gamma = \kappa_T = \kappa_S = 1$, $A = 0.5$, $p = 0.5$, $\zeta = 0.5$, and $[\Delta u'(q)]^2 = 0.4$.

These effects could well have an impact on voter learning and, more importantly, on how dispersed the final beliefs will be. There exists a non-linear relation between platform penetration and transmission quality of Social Media (see panel (a) of figure 14). For low levels of Social Media viewership, the incentives to rely more on Traditional Media's content precision are greater. After some threshold the optimal ω decreases, given that the platform is already able to send a sufficiently precise signal to its audience about ϵ_i; given that the precision of this individual signal is increasing in p. Since the platform values further engagement as well through costly learning it will decrease ω^*, since it does not want to incentivize more precision from Traditional Media.

We saw that greater platform penetration tends to reduce optimal precision by Social Media. As the right panel of figure 14 shows, even though the larger ω^* will increase precision, this seems to be outweighed by the negative impact of rising p on ψ^*. The combination of these two forces mean that higher platform penetration tends to leave Traditional Media voters with greater interim uncertainty, and thus increase their costly attention allocation.

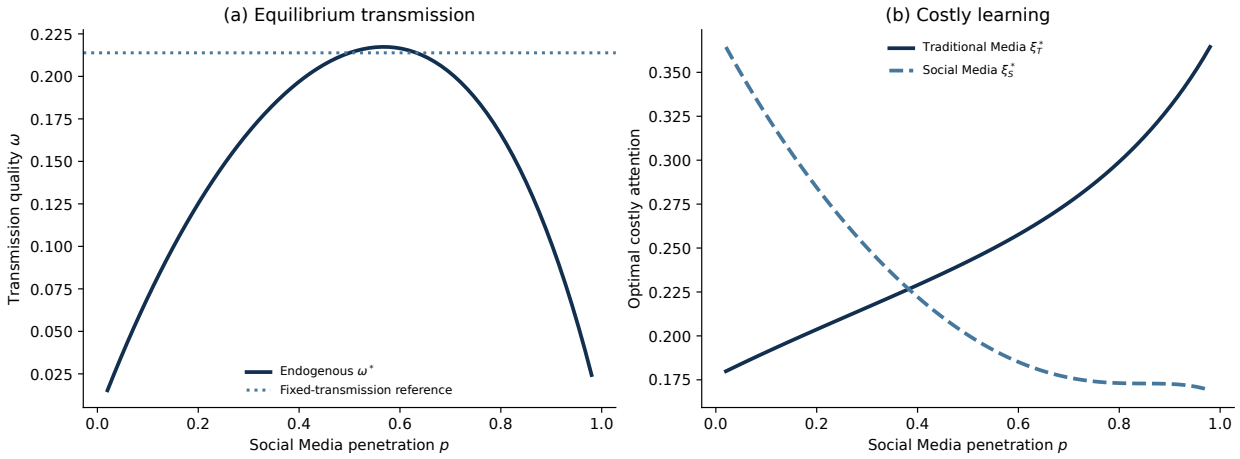


Figure 14: Equilibrium transmission and voters' final information as platform penetration varies. Panel (a) reports ω^* and the fixed-transmission reference; panel (b) reports costly attention of the two media-user groups, ξ_T^* and ξ_S^*. In the depicted interior-attention region, the remaining individual posterior variances coincide at $\chi_T = \chi_S = 1.25$. The calibration is $\tau_\mu = \tau_\epsilon = r = c = \gamma = \kappa_T = \kappa_S = 1$, $A = 0.5$, $\lambda = 0.5$, $\zeta = 0.5$, $[\Delta u'(q)]^2 = 0.4$; only p varies.

The differences in learning would impact what the ultimate dispersion of the cross-voter belief distribution would be. The technological restrictions that Traditional Media has act as a *cohesive* factor for voters' beliefs. Note that every Traditional Media voter learns about the common component. Thus, there could be situations in which the distribution of final

beliefs are compressed, or are degenerate, around a point. On the other hand, as Social Media voters reduce more uncertainty they become **more different** due to they learning better about their idiosyncratic component of policy stance. We will discuss this in more detail later.

Finally, figure 15 compares Traditional Media's maximized profits under monopoly with those obtained as Social Media penetration increases, considering both exogenous and endogenous transmission quality. In the baseline calibration, Traditional Media's profits decline with Social Media penetration under both specifications. As the share of voters consuming news through Social Media increases, Traditional Media loses direct audience revenues and receives only a fraction λ of the revenues generated through the platform. Although Traditional Media adjusts its reporting precision and saves on production costs, these adjustments do not fully compensate for the loss of revenues.

The decline is more pronounced when transmission quality is endogenous. Social Media optimally chooses its transmission quality to maximize its own profits, rather than those of the professional information producer. In the baseline calibration, this results in lower transmission quality than under the exogenous benchmark, further reducing Traditional Media's returns from information distributed through the platform. These findings illustrate how Social Media intermediation can reduce the profitability of professional news production even in the absence of competing professional information producers. The result is specific to the calibration, as sufficiently high revenues from Social Media distribution could potentially reverse this effect.

To understand the effects of Social Media penetration on Traditional Media's profitability, consider first the case of exogenous transmission quality. Applying the envelope theorem to Traditional Media's profit-maximization problem yields

$$\frac{d\Pi_T^*}{dp} = \underbrace{\lambda R_S - R_T}_{\text{Audience redistribution}} + \underbrace{p\lambda \frac{\partial R_S}{\partial p}}_{\text{Personalization effect}} . \tag{6.7}$$

The first term captures the substitution of direct audience revenues with the fraction of Social Media revenues accruing to Traditional Media. The second captures the effect of penetration on Social Media's personalization and, consequently, on voters' attention. Traditional Media's profits decrease with penetration whenever the revenues recovered through Social Media, including the personalization effect, are insufficient to compensate for the loss of direct revenues.

With endogenous transmission, Social Media also adjusts its transmission quality in re-

sponse to penetration. The corresponding expression becomes

$$\frac{d\Pi_T^*}{dp} = \lambda R_S - R_T + p\lambda \frac{\partial R_S}{\partial p} + \underbrace{p\lambda \frac{\partial R_S}{\partial \omega} \frac{\partial \omega^*}{\partial p}}_{\text{Transmission response}}, \tag{6.8}$$

where the partial derivative of ω^* with respect to p holds reporting precision fixed. Thus, endogenous transmission introduces an additional channel through which Social Media penetration influences Traditional Media's profitability.

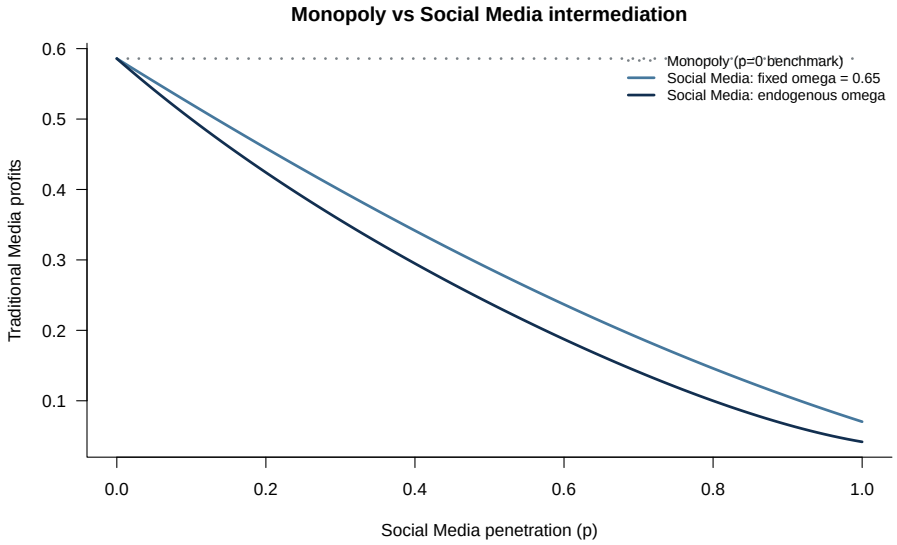


Figure 15: Traditional Media profits and Social Media penetration

Notes: The figure compares Traditional Media's maximized profits under monopoly (no Social Media) with profits under Social Media penetration, $p \in [0, 1]$. The monopoly benchmark is constant. The other two curves correspond to exogenous transmission quality ($\omega = 0.65$) and endogenous transmission quality, respectively. Traditional Media optimally chooses its reporting precision in each scenario. The revenue-sharing parameter is fixed at $\lambda = 0.5$. All profits are net of reporting costs.

Figure 15 illustrates these mechanisms. Under the baseline calibration, profits decline with penetration in both specifications, with a larger decline when transmission quality is endogenous. Although Traditional Media adjusts its reporting precision and production costs, these adjustments do not fully compensate for the loss of direct revenues. The results highlight how Social Media intermediation can reduce professional news producers' profitability even without competition in the production of professional information.

7 Equilibrium Belief Dispersion

Along this paper we have one exogenous distance on political parties' policies (the policy stakes). We know introduce what we consider our measure of polarization on the voters' side. For this purpose we focus on the dispersion in posterior final beliefs. Recall that, after the costly signal is sent, voters update their interim beliefs towards $\bar{\theta}_i$.

One rather mechanical way in which Social Media fuels dispersion in voters' beliefs is by allowing viewers to learn about individual components; whereas, as discussed above, Traditional media is unable to do so. This creates a tension between a force compressing beliefs towards a common center (the common component) and others driving them apart (the idiosyncratic ϵ_i). To focus on the dispersion brought upon by what both media outlets can report about, we will focus on a measure of dispersion of beliefs conditional on y_T, the report of Traditional Media. Naturally, we have seen that Social Media can choose to report this with less precision, which is true for any $\omega < 1$. This will have an effect on how different the final means will be and in the relative dispersion across voters.

We measure political belief polarization by the dispersion of voters' final posterior beliefs, conditional on the original report y_T produced by Traditional Media. Formally, we define

$$\mathcal{P}(y_T) = \mathrm{Var}(\bar{\theta}_{i,k} \mid y_T). \tag{7.1}$$

Where $\bar{\theta}_i$ is the posterior mean after getting the second signal. Then $\bar{\theta}_{i,k}$ is equivalent to $m_{2,i,k} = \mathbb{E}[\theta_i \mid y_{i,k}, d_{i,k}]$. Since a fraction $1-p$ of voters receives information from Traditional Media and a fraction p from Social Media, the law of total variance allows us to decompose this measure into dispersion within each audience and differences in average beliefs across audiences:

$$\mathcal{P}(y_T) = (1-p)\,\mathrm{Var}(\bar{\theta}_{i,T} \mid y_T) + p\,\mathrm{Var}(\bar{\theta}_{i,S} \mid y_T) + p(1-p)\left(\mathbb{E}[\bar{\theta}_{i,T} \mid y_T] - \mathbb{E}[\bar{\theta}_{i,S} \mid y_T]\right)^2. \tag{7.2}$$

The first two terms capture the dispersion of posterior beliefs among voters receiving information from the same outlet. The last term captures differences in average beliefs between Traditional and Social Media viewers. These differences arise because Social Media may transmit professional information with lower precision than Traditional Media.

Using the posterior updating rules derived above, the difference between the two audiences' average beliefs, conditional on y_T, is

$$\mathbb{E}[\bar{\theta}_{i,T} \mid y_T] - \mathbb{E}[\bar{\theta}_{i,S} \mid y_T] = (1-\xi_S^*)\left(\frac{\psi}{\tau_\mu + \psi} - \frac{\omega\psi}{\tau_\mu + \omega\psi}\right) y_T. \tag{7.3}$$

Hence, the difference between average beliefs disappears when $\omega = 1$, so that Social Media transmits professional information without distortion, or when $\xi_S^* = 1$, so that Social Media voters acquire fully revealing additional information.

Both within-audience dispersion and differences between audiences depend on the information provided by media outlets and voters' subsequent attention decisions. In particular, greater costly attention weakly increases the conditional dispersion of posterior beliefs among Traditional Media viewers. For Social Media viewers, the relationship also depends on transmission quality ω.

Since $\mathcal{P}(y_T)$ depends on the realization of the original report, we can average over its distribution to obtain a measure of expected conditional belief dispersion.

Proposition 5 (Dispersion of voters' beliefs)**.** *The expected conditional dispersion of voters' final posterior beliefs is*

$$\begin{aligned}\overline{\mathcal{P}} = (1-p)\mathbb{E}_{y_T}\left[\mathrm{Var}(\bar{\theta}_{i,T} \mid y_T)\right] + p\mathbb{E}_{y_T}\left[\mathrm{Var}(\bar{\theta}_{i,S} \mid y_T)\right] \\ + p(1-p)(1-\xi_S^*)^2\left(\frac{\psi}{\tau_\mu+\psi} - \frac{\omega\psi}{\tau_\mu+\omega\psi}\right)^2 \mathrm{Var}(y_T).\end{aligned} \tag{7.4}$$

Expression (7.4) highlights two channels through which media decisions shape political belief polarization. First, reporting precision and costly attention affect the dispersion of beliefs within each audience. Second, imperfect transmission of professional information can generate differences between the average beliefs of Traditional and Social Media viewers. The overall effect depends on the equilibrium reporting and transmission decisions of both outlets.

The derivation is provided in Appendix A.5.

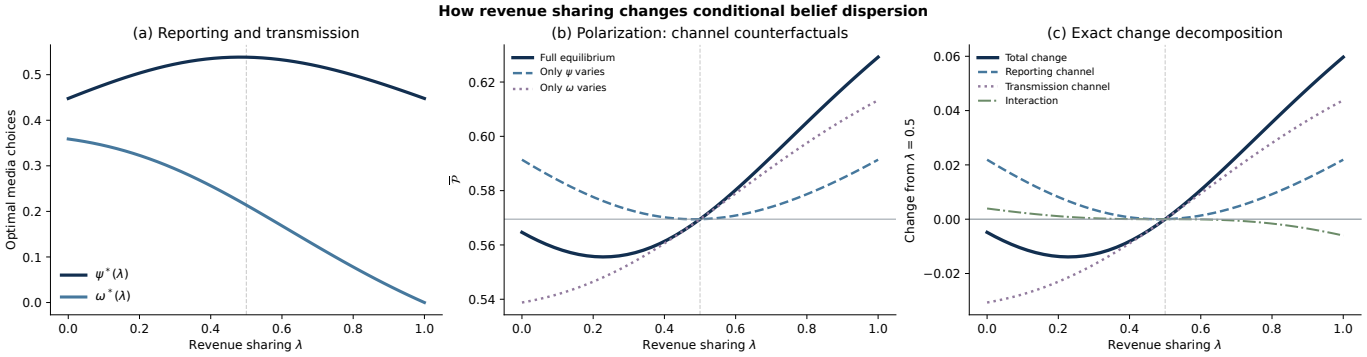


Figure 16: Effects of changes in revenue sharing parameter λ on the dispersion of beliefs.

In figure 16 we can disentangle the channels that drive (7.4). In panel (a) we see the mechanism that was already explained in figure 13 above. The effect of greater revenue share

is concave on optimal precision of Traditional Media report, whereas the chosen transmission of Social Media, ω^*, decreases monotonically on λ. Panels (b) and (c) shows us how this impacts voter polarization. A first force driving polarization down is that for low λ Traditional Media is increasing its precision, which delivers a lower $\mathcal{D}_T$. Although a lower ω might be driving more polarization through relatively poorer common information by Social Media voters, the cohesive force of better information about the common component more than dominates this. Then, after some point the relative magnitudes of these effects reverse and polarization starts increasing, as λ increases above 0.5, Traditional Media decreases its reporting quality and both the effects of simultaneously decreasing ψ and ω result in higher polarization. At this range, precision of Traditional Media report is relatively low, while this is compounded by Social Media sending relatively worse quality information about the common component. Both these effects drive up ξ_T and ξ_S.

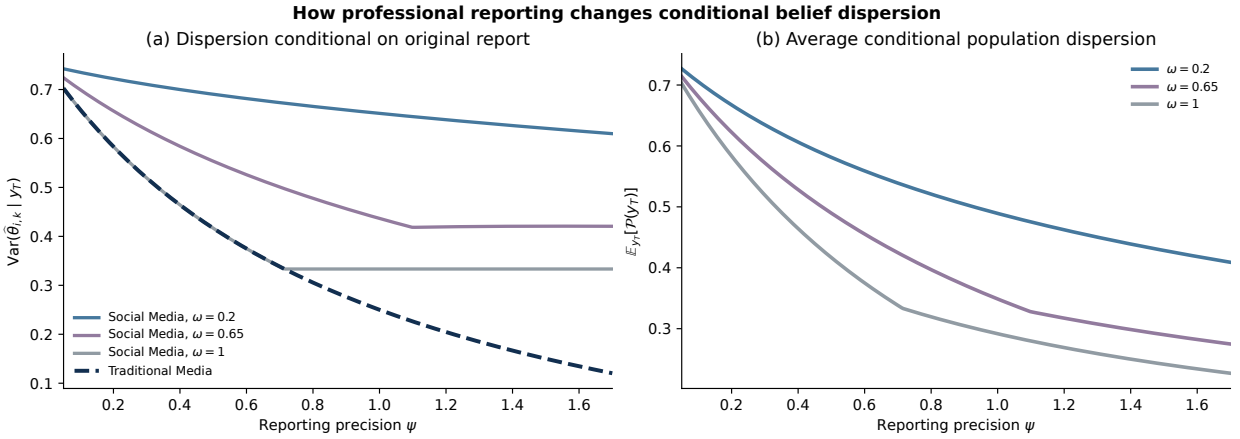


Figure 17: Professional reporting and conditional belief dispersion. Panel (a) shows the conditional variance of final posterior expected policy positions for Traditional Media and Social Media viewers. Panel (b) reports expected population polarization, incorporating within-group dispersion and differences between the groups' conditional average beliefs.

The effects of Traditional Media's reporting decisions on belief dispersion can be examined more directly in Figure 17. In both panels, we hold Social Media's transmission quality, ω, fixed and vary professional reporting precision, ψ, allowing voters to adjust their optimal costly attention accordingly. Panel (a) presents the conditional dispersion of final posterior means for each group, whereas panel (b) reports the corresponding aggregate polarization measure.

In the illustrated calibration, improvements in reporting precision reduce aggregate conditional belief dispersion for every transmission quality considered. Greater precision about the common component reduces the uncertainty remaining after the initial Traditional Media

report and changes voters' incentives to acquire additional information. In particular, when costly attention is interior, better reporting decreases the conditional dispersion of posterior means within each group.

The magnitude of this effect depends on the transmission quality chosen by Social Media. When Social Media transmits in a better way the Traditional Media's report (higher ω), improvements in reporting precision translate more effectively into reductions in Social Media viewers' conditional belief dispersion. As reporting becomes sufficiently precise, these viewers may cease acquiring additional costly information, so that subsequent changes in their beliefs are determined exclusively by their initial common and personalized signals.

An important distinction arises at this point. Under $\omega = 1$, once Social Media's costly attention reaches zero, its conditional belief dispersion becomes constant and reflects only the information revealed about heterogeneous individual preferences. Under imperfect transmission, however, reporting precision can continue affecting conditional dispersion even without additional costly learning, because the curated professional report remains a noisy transformation of the original signal. Moreover, aggregate polarization may continue changing through the dispersion of Traditional Media viewers' beliefs and differences between the groups' conditional average posterior positions.

Lastly, we can see how these differences show up when we plot the entire distributions of final beliefs in figure 18. We see in panel (a) that the posterior mean of Traditional Media voters is closer to the report, while it exhibits less dispersion than the distribution of Social Media viewers. In this sense, Social Media voters are more polarized than Traditional media voters. On the other hand, the middle panel shows an interesting case in which both distributions are exactly the same. No voter population is more polarized than the other. The final case in panel (c) shows a situation in which the distribution of Social Media voters' beliefs is a mean-preserving spread of the one of Traditional Media. Although both distributions have the same mean, Social Media voters have more dispersed beliefs. Thus, they are more polarized. These effects speak more clearly towards within-group polarization than our previous discussions.

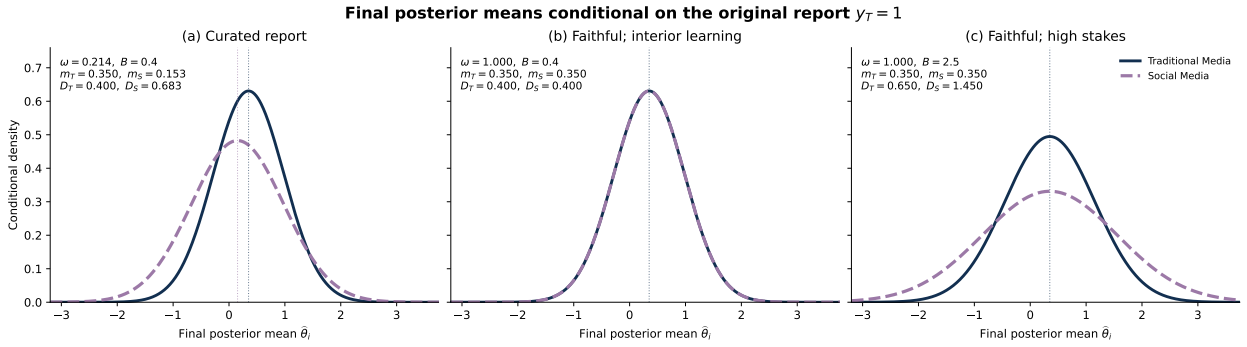


Figure 18: Simulated distributions of final beliefs for different cases. The original report by Traditional Media is fixed at $y_T = 1$. In these plots B stands for the political stakes. Panel (a) shows a situation with $\omega < 1$, to induce different posterior means. Panel (b) and (c) have $\omega = 1$. The latter shows the impact of having high political stakes, which induces higher costly learning.

8 Conclusions

This paper studies how the expansion of Social Media affects professional news production, voters' information acquisition, and political belief polarization. We develop a model in which Traditional Media produces information about a common policy-relevant state, while Social Media intermediates this content and combines it with personalized information. Voters are rationally inattentive and decide how much additional information to acquire after receiving an initial signal.

Our analysis highlights three main findings. First, greater Social Media penetration can weaken Traditional Media's incentives to invest in professional reporting. As a larger share of voters obtain information through Social Media, the returns to reporting quality increasingly depend on the compensation received from the platform rather than directly from viewers. In our baseline calibration, greater Social Media penetration reduces both reporting precision and Traditional Media's profits, with the decline in profitability being more pronounced when transmission quality is endogenous. These results illustrate how digital intermediation can weaken the incentives and profitability of professional information production even without competition among original news producers.

Second, allowing Social Media to choose how effectively it transmits professional information introduces an additional strategic interaction between the two outlets. Revenue sharing encourages Traditional Media to invest in reporting quality, but also reduces the share of revenues retained by Social Media, potentially weakening its incentives to transmit that in-

formation. Our numerical results show that this interaction can generate non-monotonic effects of revenue sharing on professional reporting quality.

Third, these decisions have implications for political belief polarization. Social Media provides information about voters' idiosyncratic preferences, while Traditional Media primarily informs them about a common component. Changes in reporting precision and transmission quality affect both the dispersion of beliefs within each audience and differences in average beliefs across audiences. In the example we have showed, it is clear how revenue-sharing arrangements can generate rich effects on expected conditional belief dispersion.

Taken together, our findings suggest that the effects of Social Media on political information cannot be understood solely through the content voters consume. The economic relationship between original information producers and digital intermediaries also shapes how much information is produced, how effectively it is transmitted, and how voters ultimately form their political beliefs.

Several extensions could build on this framework. One direction would be to endogenize voters' choice of media outlet, allowing them to select between Traditional and Social Media according to the information available and the costs of acquiring it. This would introduce an additional feedback between reporting quality, platform transmission, and audience composition.

A second extension would be to endogenize the revenue-sharing arrangement between the two media outlets. Rather than treating the compensation received by Traditional Media as given, a bargaining or pricing mechanism could determine how the value generated by professional information is divided between the producer and the intermediary. This would allow us to examine how different contractual arrangements or regulatory interventions affect information production and transmission.

Finally, the framework could be extended to study endogenous political competition. Allowing parties to choose their policy positions in response to voters' posterior beliefs would help establish whether changes in the information environment translate into greater policy polarization, rather than only into greater dispersion of voters' beliefs.

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A Proofs

A.1 Proof of Lemma 1

Case 1: Zero attention. If $\chi_T \geq V_{1,T}$, then $\xi_T^* = 0$, so $\frac{\partial \xi_T^*}{\partial \psi} = 0$.
Case 2: Interior attention. If $b < \chi_T < V_{1,T}$, $\xi_T^* = 1 - \frac{\chi_T}{a(\psi)+b}$. Differentiating,

$$\frac{\partial \xi_T^*}{\partial \psi} = \frac{\chi_T a'(\psi)}{[a(\psi)+b]^2} = -\frac{\chi_T}{[1+b(\tau_\mu+\psi)]^2} < 0.$$

Case 3: Attention ceiling binds: If $\chi_T \leq b$, $\xi_T^* = \bar{\xi}_T = \frac{a(\psi)}{a(\psi)+b}$. Therefore,

$$\frac{\partial \xi_T^*}{\partial \psi} = \frac{b\, a'(\psi)}{[a(\psi)+b]^2} = -\frac{b}{[1+b(\tau_\mu+\psi)]^2} < 0.$$

Thus, combining all three cases, $\frac{\partial \xi_T^*}{\partial \psi} \leq 0$.

A.2 Proof of Proposition 2

Take the additive revenue function of the profits for ψ and ξ and maximize with respect to ψ:

$$\Pi_T(\psi) = \underbrace{(1-p)R_T\left(\psi, \xi_T(\psi)\right)}_{\text{Direct users}} + \underbrace{p\lambda R_S(\psi, \xi_S(\psi); \omega)}_{\text{From Social Media}} - C_T(\psi). \tag{A.1}$$

Then the FOC is straightforward, this delivers an interior ψ by

$$\psi \in \operatorname{argmax} \left\{ (1-p)R_T\left(\psi, \xi_T(\psi)\right) + p\lambda R_S(\psi, \xi_S(\psi); \omega) - C_T(\psi) \right\}$$

Here social media plays a role by being the follower and take as given the optimal ψ, so T will get part of its profits weighted by λ.

A.3 Proof for Proposition 3

Proof. Since Traditional Media optimally chooses its reporting precision following entry, its equilibrium profits are at least those obtained by choosing the monopoly-optimal precision ψ^M. Evaluating both profit functions at this precision gives

$$\Pi_T(\psi^M, \omega) - \Pi_T^M(\psi^M) = p\left[\lambda R_S(\psi^M, \xi_S^*; \omega) - R_T(\psi^M, \xi_T^*)\right].$$

The reporting cost $c(\psi^M)^2/2$ cancels because the same precision is evaluated in both cases. Under the stated condition, the difference is nonnegative. Since Traditional Media can reoptimize its reporting precision after entry, its maximized profits must be at least as high as its monopoly profits.

A.4 Proof for Proposition 4

Take the following maximization problem for S media.

$$\omega^*(\psi) \in \arg\max_{\omega\in[0,1]} \left\{ p(1-\lambda)\left[r\omega\psi + A\xi_S^*(\psi,\omega,p)\right] - \frac{\zeta}{2}\omega^2 \right\}. \tag{A.2}$$

The subgame-perfect equilibrium is a pair (ψ^*, ω^*) such that T media anticipates the best-response of S media once ψ has been chosen. Then taking FOC we have the interior solution we have a fixed point solution for ω shown in the Proposition.

A.5 Proof of Belief Dispersion

This contains the proof for Proposition 5.

In here we derive in detail the dispersion of beliefs measure which constitutes our main statistic for voter polarization. Using $\mathrm{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$, the cross-sectional variance of final beliefs in the population is

$$\mathrm{Var}_i\left(\bar{\theta}_i \mid y_T\right) = (1-p)\mathbb{E}_i\left[\bar{\theta}_{i,T}^2 \mid y_T\right] + p\mathbb{E}_i\left[\bar{\theta}_{i,S}^2 \mid y_T\right] - \left[(1-p)\mathbb{E}_i\left[\bar{\theta}_{i,T} \mid y_T\right] + p\mathbb{E}_i\left[\bar{\theta}_{i,S} \mid y_T\right]\right]^2. \tag{A.3}$$

Using $\mathbb{E}[X^2] = \mathrm{Var}(X) + \mathbb{E}[X]^2$, we obtain

$$\begin{aligned}\mathrm{Var}_i\left(\bar{\theta}_i \mid y_T\right) = {} & (1-p)\,\mathrm{Var}_i\left(\bar{\theta}_{i,T} \mid y_T\right) + p\,\mathrm{Var}_i\left(\bar{\theta}_{i,S} \mid y_T\right) \\ & + (1-p)\left[\mathbb{E}_i\left(\bar{\theta}_{i,T} \mid y_T\right)\right]^2 + p\left[\mathbb{E}_i\left(\bar{\theta}_{i,S} \mid y_T\right)\right]^2 \\ & - \left[(1-p)\mathbb{E}_i\left(\bar{\theta}_{i,T} \mid y_T\right) + p\mathbb{E}_i\left(\bar{\theta}_{i,S} \mid y_T\right)\right]^2. \end{aligned} \tag{A.4}$$

Expanding the last squared term and collecting terms gives

$$\begin{aligned}\mathrm{Var}_i\left(\bar{\theta}_i \mid y_T\right) = {} & (1-p)\,\mathrm{Var}_i\left(\bar{\theta}_{i,T} \mid y_T\right) + p\,\mathrm{Var}_i\left(\bar{\theta}_{i,S} \mid y_T\right) \\ & + p(1-p)\left[\mathbb{E}_i\left(\bar{\theta}_{i,S} \mid y_T\right) - \mathbb{E}_i\left(\bar{\theta}_{i,T} \mid y_T\right)\right]^2. \end{aligned} \tag{A.5}$$

The last term is the between-platform component of belief dispersion. More generally, between-group variance is obtained by weighting the squared distance between each group

mean and the population mean. In our two-platform setting, this is

$$(1-p)\left[\mathbb{E}_i\left(\bar{\theta}_{i,T} \mid y_T\right) - \mathbb{E}_i\left(\bar{\theta}_i \mid y_T\right)\right]^2 + p\left[\mathbb{E}_i\left(\bar{\theta}_{i,S} \mid y_T\right) - \mathbb{E}_i\left(\bar{\theta}_i \mid y_T\right)\right]^2. \tag{A.6}$$

Since

$$\mathbb{E}_i\left[\bar{\theta}_i \mid y_T\right] = (1-p)\mathbb{E}_i\left[\bar{\theta}_{i,T} \mid y_T\right] + p\mathbb{E}_i\left[\bar{\theta}_{i,S} \mid y_T\right],$$

we have

$$\mathbb{E}_i\left[\bar{\theta}_{i,T} \mid y_T\right] - \mathbb{E}_i\left[\bar{\theta}_i \mid y_T\right] = p\left[\mathbb{E}_i\left(\bar{\theta}_{i,T} \mid y_T\right) - \mathbb{E}_i\left(\bar{\theta}_{i,S} \mid y_T\right)\right], \tag{A.7}$$

and

$$\mathbb{E}_i\left[\bar{\theta}_{i,S} \mid y_T\right] - \mathbb{E}_i\left[\bar{\theta}_i \mid y_T\right] = (1-p)\left[\mathbb{E}_i\left(\bar{\theta}_{i,S} \mid y_T\right) - \mathbb{E}_i\left(\bar{\theta}_{i,T} \mid y_T\right)\right]. \tag{A.8}$$

Substituting these expressions into (A.6),

$$\begin{aligned}
&(1-p)p^2\left[\mathbb{E}_i\left(\bar{\theta}_{i,S} \mid y_T\right) - \mathbb{E}_i\left(\bar{\theta}_{i,T} \mid y_T\right)\right]^2 \\
&\quad + p(1-p)^2\left[\mathbb{E}_i\left(\bar{\theta}_{i,S} \mid y_T\right) - \mathbb{E}_i\left(\bar{\theta}_{i,T} \mid y_T\right)\right]^2 \\
&= p(1-p)\left[p + (1-p)\right]\left[\mathbb{E}_i\left(\bar{\theta}_{i,S} \mid y_T\right) - \mathbb{E}_i\left(\bar{\theta}_{i,T} \mid y_T\right)\right]^2 \\
&= p(1-p)\left[\mathbb{E}_i\left(\bar{\theta}_{i,S} \mid y_T\right) - \mathbb{E}_i\left(\bar{\theta}_{i,T} \mid y_T\right)\right]^2.
\end{aligned} \tag{A.9}$$

Thus, the squared difference in conditional means in (A.5) is simply the two-group version of the general between-group variance.

We now compute the conditional expectations of voters' final posterior beliefs, which determine the differences between the two media audiences.

Given our assumption that the prior mean is $m_0 = 0$, the average final belief of Traditional Media viewers, conditional on the original report y_T, is

$$\mathbb{E}[\bar{\theta}_{i,T} \mid y_T] = \frac{\psi}{\tau_\mu + \psi} y_T. \tag{A.10}$$

For Social Media viewers, we need to account for the fact that the platform may distort the initial signal about the common component through its transmission quality ω. However, as assumed in the main text, this distortion does not affect the subsequent costly signal.

After receiving the costless signal, a Social Media voter forms the following interim posterior mean and variance:

$$m_{1,S} = \frac{\omega\psi}{\tau_\mu + \omega\psi}\hat{y}_T + \frac{\gamma p}{\tau_\epsilon + \gamma p}\hat{\epsilon}_i, \qquad V_{1,S} = \frac{1}{\tau_\mu + \omega\psi} + \frac{1}{\tau_\epsilon + \gamma p}.$$

After acquiring the additional costly signal $d_{i,S}$, the final posterior belief can be expressed in terms of optimal attention:

$$\bar{\theta}_{i,S} = (1 - \xi_S^*)m_{1,S} + \xi_S^* d_{i,S}. \tag{A.11}$$

Taking conditional expectations with respect to the original report y_T, and using the independence of the signal errors, we obtain

$$\mathbb{E}[d_{i,S} \mid y_T] = \mathbb{E}[\mu \mid y_T] = \frac{\psi}{\tau_\mu + \psi} y_T,$$

and

$$\mathbb{E}[m_{1,S} \mid y_T] = \frac{\omega\psi}{\tau_\mu + \omega\psi} y_T.$$

Substituting these expressions into (A.11), the conditional average final belief of Social Media viewers is

$$\mathbb{E}[\bar{\theta}_{i,S} \mid y_T] = \left[\xi_S^* \frac{\psi}{\tau_\mu + \psi} + (1 - \xi_S^*)\frac{\omega\psi}{\tau_\mu + \omega\psi}\right] y_T. \tag{A.12}$$

Combining (A.10) and (A.12), the difference between the two audiences' conditional average beliefs becomes

$$\mathbb{E}[\bar{\theta}_{i,T} \mid y_T] - \mathbb{E}[\bar{\theta}_{i,S} \mid y_T] = (1 - \xi_S^*)\left(\frac{\psi}{\tau_\mu + \psi} - \frac{\omega\psi}{\tau_\mu + \omega\psi}\right) y_T. \tag{A.13}$$

This difference disappears when $\omega = 1$ or when Social Media voters acquire fully revealing additional information.

We now turn to the dispersion of posterior beliefs within each audience. By the law of total variance, the expected conditional variance of final posterior means is equal to the prior variance of θ_i, minus the remaining posterior variance and the variance of the conditional average belief.

For Traditional Media viewers, this gives

$$\mathbb{E}_{y_T}\left[\mathrm{Var}(\bar{\theta}_{i,T} \mid y_T)\right] = V_{1,T} - V_{2,T} = \left(\frac{1}{\tau_\mu + \psi} + \frac{1}{\tau_\epsilon}\right)\xi_T^*. \tag{A.14}$$

For Social Media viewers, the corresponding expression is

$$\mathbb{E}_{y_T}\left[\text{Var}(\bar{\theta}_{i,S} \mid y_T)\right] = V_0 - V_{2,S} - \left[\xi_S^* \frac{\psi}{\tau_\mu + \psi} + (1 - \xi_S^*)\frac{\omega\psi}{\tau_\mu + \omega\psi}\right]^2 \text{Var}(y_T), \tag{A.15}$$

where V_0 is the prior variance of θ_i and $V_{2,S} = V_{1,S}(1 - \xi_S^*)$.

Finally, substituting (A.13), (A.14), and (A.15) into the law-of-total-variance decomposition in (7.2), and taking expectations over y_T, gives

$$\begin{aligned}\overline{\mathcal{P}} = (1-p)\mathbb{E}_{y_T}\left[\text{Var}(\bar{\theta}_{i,T} \mid y_T)\right] + p\mathbb{E}_{y_T}\left[\text{Var}(\bar{\theta}_{i,S} \mid y_T)\right] \\ + p(1-p)(1-\xi_S^*)^2\left(\frac{\psi}{\tau_\mu + \psi} - \frac{\omega\psi}{\tau_\mu + \omega\psi}\right)^2 \text{Var}(y_T).\end{aligned} \tag{A.16}$$

This establishes Proposition 5.

B Political Competition

We have assumed political competition to be as given in the main text. However, politicians plays a role in the model by choosing their optimal policies. This follows a probabilistic voting model and will affect information acquisition by voters as in Matějka and Tabellini (2021).

We examine how voters' information acquisition affects the policies proposed by political parties. Consider two parties, $j \in \{L, R\}$, choosing policies q_L and q_R. As in the previous sections, voter i has an underlying preferred policy $\theta_i = \mu + \epsilon_i$, which is imperfectly known.

After acquiring information from outlet $k \in \{T, S\}$, voters form final posterior beliefs $\bar{\theta}_{i,k}$. Given these beliefs, their expected utility from the policy proposed by party j is

$$\mathbb{E}_i[U_i(q_j)] = -(q_j - \bar{\theta}_{i,k})^2 - V_{2,k}. \tag{B.1}$$

After forming their posterior beliefs, voters receive a valence or popularity shock η_i in favor of party R. This shock consists of an idiosyncratic component, which differs across voters, and an aggregate component, which is common to the entire electorate. Formally, $\eta_i = \tilde{\eta}_i + \bar{\eta}$, where

$$\tilde{\eta}_i \sim \mathcal{U}\left(-\frac{1}{2\phi}, \frac{1}{2\phi}\right), \qquad \bar{\eta} \sim \mathcal{U}\left(-\frac{1}{2\Psi}, \frac{1}{2\Psi}\right). \tag{B.2}$$

The parameters $\phi > 0$ and $\Psi > 0$ denote the densities of the idiosyncratic and aggregate valence shocks, respectively. We assume that both components are independent of each other and of voters' policy preferences and informational signals.

The idiosyncratic shock introduces heterogeneity in voters' preferences over candidates beyond their policy positions, while the aggregate shock introduces uncertainty about the electoral outcome that is common to all voters.

Voter i supports party L whenever the expected utility from its proposed policy exceeds that from party R, including the valence advantage of the latter:

$$-(q_L - \bar{\theta}_{i,k})^2 \geq -(q_R - \bar{\theta}_{i,k})^2 + \eta_i. \tag{B.3}$$

Equivalently, voter i supports party L whenever

$$\tilde{\eta}_i \leq (q_R - \bar{\theta}_{i,k})^2 - (q_L - \bar{\theta}_{i,k})^2 - \bar{\eta}. \tag{B.4}$$

Thus, voting decisions depend on both the posterior beliefs formed through information acquisition and the realization of valence shocks.

For any given realization of the valence shocks, voters compare the utilities associated with the two proposed policies. When valence shocks are absent, preferences over policy are single-peaked and the Median Voter Theorem applies directly.

In that case, the relevant median is determined by the distribution of final posterior beliefs. Since a fraction $1 - p$ of voters receives information from Traditional Media and a fraction p from Social Media, the median posterior belief $\bar{\theta}^{med}$ satisfies

$$(1 - p)\Pr(\bar{\theta}_{i,T} \leq \bar{\theta}^{med}) + p\Pr(\bar{\theta}_{i,S} \leq \bar{\theta}^{med}) = \frac{1}{2}. \tag{B.5}$$

For two office-motivated parties, the standard Median Voter Theorem implies convergence to this position:

$$q_L^* = q_R^* = \bar{\theta}^{med}. \tag{B.6}$$

With valence shocks, however, voting decisions are no longer determined exclusively by voters' posterior ideal points. The aggregate shock introduces uncertainty about electoral outcomes, while idiosyncratic shocks introduce additional heterogeneity in party support. Consequently, the standard median-voter convergence result does not generally characterize the equilibrium of the probabilistic-voting model.

Nevertheless, the distribution of posterior beliefs remains central to political competition, since media decisions affect the policy component of voters' electoral preferences.

Our framework distinguishes between the dispersion of voters' posterior beliefs and the policy positions chosen by political parties. Changes in Traditional Media's reporting precision ψ and Social Media's transmission quality ω affect the distribution of $\bar{\theta}_{i,k}$ and therefore

the policy preferences relevant for electoral competition.

However, greater dispersion of posterior beliefs does not necessarily imply greater divergence between parties' policies. Under the standard Median Voter Theorem, parties converge to the median even when the distribution becomes more dispersed. With valence shocks, electoral outcomes additionally depend on the realization of popularity shocks.

For the main analysis, we maintain exogenous party positions, allowing political stakes to affect voters' rational-inattention decisions. The present extension illustrates how the resulting distribution of posterior beliefs may subsequently influence electoral competition.

C Alternative Revenue Specification: Multiplicative Revenues

In the main analysis, we assume that reporting quality and voter attention contribute additively to media revenues. In this appendix, we consider an alternative specification in which these two components are complementary. In particular, revenues are generated by the interaction between the quality of the information provided and the attention it receives.

We maintain the same information structure, voter learning problem, timing of the game, and production and transmission costs as in the baseline model. The only difference concerns the revenue functions, which are now given by

$$R_T(\psi, \xi_T^*) = r\psi\xi_T^*, \qquad R_S(\psi, \xi_S^*; \omega) = r\omega\psi\xi_S^*. \tag{C.1}$$

Unlike the additive specification, where reporting quality generates revenues independently of subsequent attention, the multiplicative specification requires both components to be positive. This captures an environment in which the commercial value of professional information depends on users' engagement with that information.

Under monopoly, Traditional Media chooses reporting precision to maximize

$$\Pi_T^M(\psi) = r\psi\xi_T^*(\psi) - \frac{c}{2}\psi^2. \tag{C.2}$$

For an interior optimum, the first-order condition is

$$r\left[\xi_T^* + \psi\frac{\partial\xi_T^*}{\partial\psi}\right] = c\psi. \tag{C.3}$$

This condition differs from the additive benchmark because the marginal return to reporting now depends directly on the level of attention. Although improving reporting quality

raises its commercial value, it also reduces the incentives for subsequent costly learning.

With Social Media entry and fixed transmission quality, Traditional Media instead maximizes

$$\Pi_T(\psi) = (1-p)r\psi\xi_T^* + p\lambda r\omega\psi\xi_S^* - \frac{c}{2}\psi^2. \tag{C.4}$$

The corresponding interior first-order condition is

$$(1-p)r\left[\xi_T^* + \psi\frac{\partial\xi_T^*}{\partial\psi}\right] + p\lambda r\omega\left[\xi_S^* + \psi\frac{\partial\xi_S^*}{\partial\psi}\right] = c\psi. \tag{C.5}$$

As in the additive model, Social Media penetration redistributes revenues between direct and intermediated users. However, the commercial returns from reporting are now more sensitive to voters' costly attention decisions.

When transmission quality is endogenous, Social Media chooses ω after observing Traditional Media's reporting precision. Its problem becomes

$$\max_{\omega\in[0,1]}\left\{p(1-\lambda)r\omega\psi\xi_S^* - \frac{\zeta}{2}\omega^2\right\}. \tag{C.6}$$

For an interior optimum, the first-order condition is

$$p(1-\lambda)r\psi\left[\xi_S^* + \omega\frac{\partial\xi_S^*}{\partial\omega}\right] = \zeta\omega. \tag{C.7}$$

This expression highlights the tension between transmission quality and costly attention. Greater transmission quality increases the effective precision of professional information reaching Social Media users, but reduces their incentives to acquire further information. Under multiplicative revenues, this reduction in attention also diminishes the commercial value of transmitted reporting.

Traditional Media anticipates this response when choosing reporting precision. Its equilibrium choice therefore incorporates both the direct marginal effects in equation (C.5) and the indirect effect operating through Social Media's optimal transmission decision.

When we compare with the additive specification, the main elements are as follows

We next allow Social Media to choose transmission quality endogenously. Figure 19 illustrates the equilibrium responses of reporting and transmission to changes in revenue sharing.

Table 4: Baseline Equilibrium under Alternative Revenue Specifications

Revenue specification	Reporting precision ψ^*	Transmission quality ω^*	Belief dispersion $\overline{\mathcal{P}}$
Additive	0.5384	0.2139	0.5695
Multiplicative	0.1481	0.0183	0.6984

Notes: The table reports the baseline equilibrium under additive and multiplicative revenue specifications. Reporting precision and Social Media transmission quality are endogenously determined in both specifications. Belief dispersion is measured by the expected conditional variance of voters' final posterior beliefs. All other parameters are held at their baseline values.

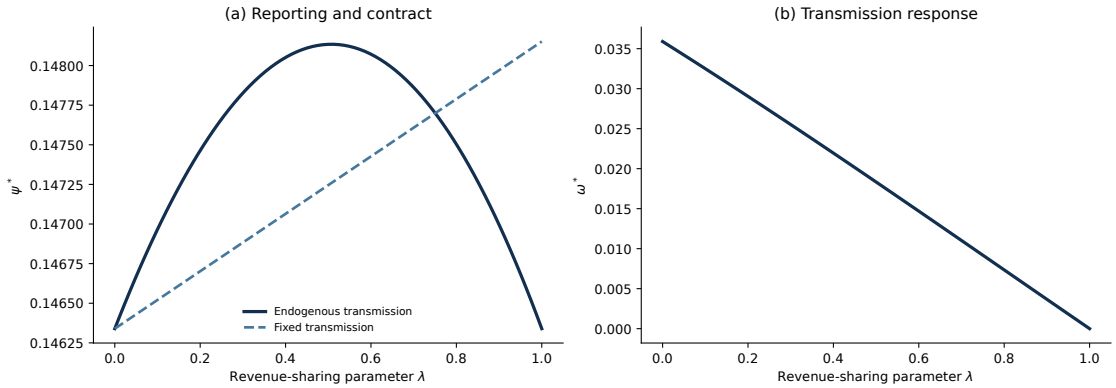


Figure 19: Revenue sharing and endogenous transmission under multiplicative revenues

Notes: Panel (a) reports Traditional Media's optimal reporting precision under endogenous and fixed transmission. Panel (b) shows Social Media's optimal transmission quality. Revenues are multiplicative, while all other assumptions remain unchanged.

Finally, figure 20 examines the implications for expected conditional belief dispersion.

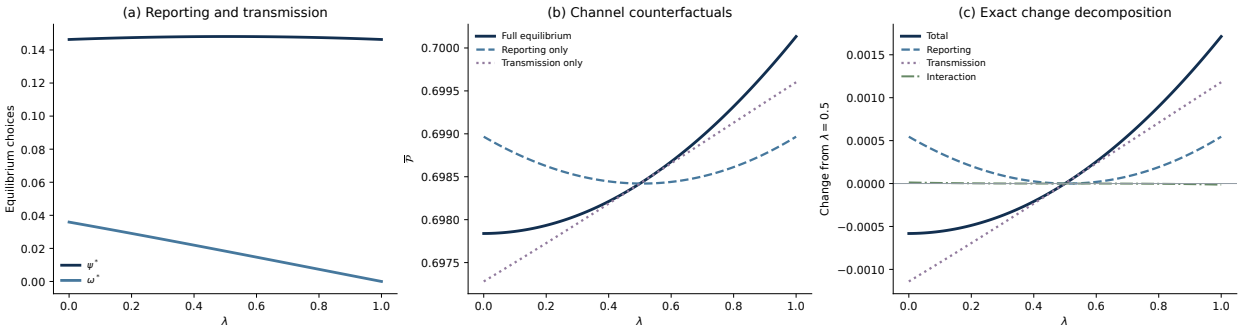


Figure 20: Revenue sharing and belief dispersion under multiplicative revenues

Notes: The figure reports equilibrium reporting and transmission decisions, expected conditional belief dispersion, and the corresponding counterfactual channels as the revenue-sharing parameter varies. The measure of belief dispersion is identical to that used in the main analysis.

The comparison illustrates that the quantitative and qualitative implications of revenue

sharing depend on how information quality and attention generate commercial revenues. While the additive specification allows reporting quality to generate revenues independently of costly attention, the multiplicative specification makes their interaction central to both media outlets' incentives. Consequently, the non-monotonic polarization response obtained in the additive benchmark should not be interpreted as a general prediction independent of the revenue specification.

D Alternative Mutual Information Costs

In our main framework we have assumed that costly information acquisition by voters means they incur a cost that is proportional to the uncertainty they reduce about the policy stance. This, as mentioned results in Traditional Media voters potentially being able to learn μ perfectly, while they remain with the same original variance on ϵ_i.

One alternative formulation we discussed is to make the mutual information cost **proportional to the elements in the costly signal**. In that case, the problem of the voters of Social Media does not suffer any alterations, since they can learn about both components of θ_i. However, making the cost proportional *only* to μ for Traditional Media voters fundamentally shape how costly is learning to them. Particularly, under this alternative, it is infinitely costly for them to learn all about μ. Thus, in this case they are unable to reach the *learning ceiling* ξ_T.

More formally, suppose that under this alternative, the mutual information cost specification is as follows for Traditional Media voters:

$$\kappa_T I(\mu; d_{i,T} | \mathcal{I}_{1,i,T}) \tag{D.1}$$

This makes it clear that the cost is proportional only to the common component. Hence, we will have a different expression for the optimal ξ_T. Particularly, for a mutual information cost given by (D.1), under normally distributed signals:

$$\xi_T^* = \max \left\{ 0, \frac{a(\psi) - \chi_T}{a(\psi) + b} \right\} \tag{D.2}$$

How these changes the incentives to exert costly learning can be better seen in the figure 21 below. There we see immediately on panel (a) that the alternative costs makes for a much lower attention allocation for Traditional Media voters for each level pf political stakes. This is due to the fact that, effectively, the alternative cost makes it for them much more costly to spend an additional unit of attention in refining the second signal. Moreover, it is

immediate to see that for a large range of values, the benchmark cost induces voters to stay at the learning ceiling, which is determined by ψ. On the other hand, we see that changing our mutual information cost has little to no impact on Social Media voters' attention allocation. In this regard, the impact comes from the interaction of changes in precision from Traditional Media and the interim uncertainty left for Social Media voters.

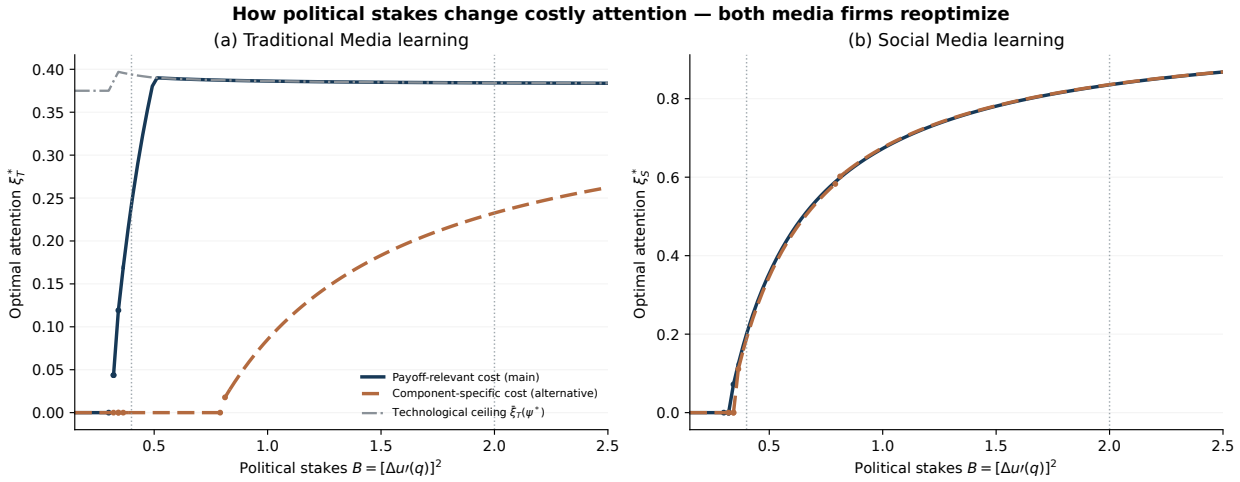


Figure 21: Costly learning allocation for Traditional and Social Media. Comparison of benchmark cost and alternative.

We can further explore how this alternative formulation changes the optimal responses of media outlets. In what follows we will assume that political stakes are high enough that under both formulations we have positive costly attention. In figure 22 we show that the qualitative responses are very similar across specifications. The effect of the revenue share parameter has the same shape than in our benchmark, while the chosen precision of reporting is lower for every value of λ. When looking at the differences in Social Media's quality of transmission, we see that the changes are less notorious. For low values of λ the optimal ω^* is just below the choices in our benchmark formulation. After some point the two curves practically overlap. Thus, our conclusions hold for the effects of the revenue share parameter on media's decisions regardless of our modeling choice for the mutual information cost.

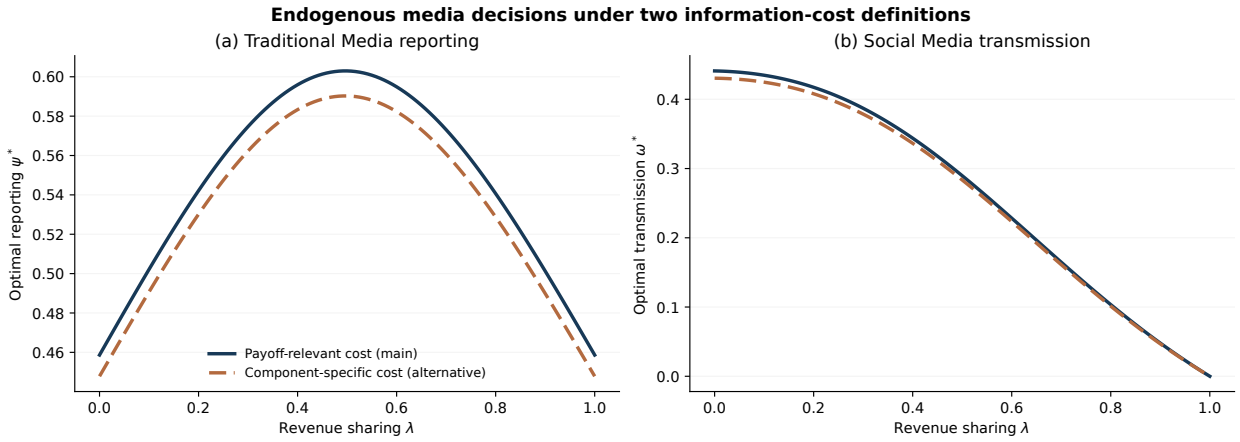


Figure 22: Effect of revenue share parameter on media's optimal choices. Benchmark vs alternative.

The effects that these changes have on equilibrium voter attention are displayed in figure 23. It basically shows that the lower costly attention for Traditional Media voters we saw in figure 21 also translates in here for every value of λ, when compared to the benchmark specification for the mutual information cost. Conversely, Social Media voters will have a weakly higher level of attention for any possible value of λ. This acts as an enhancing force of polarization.

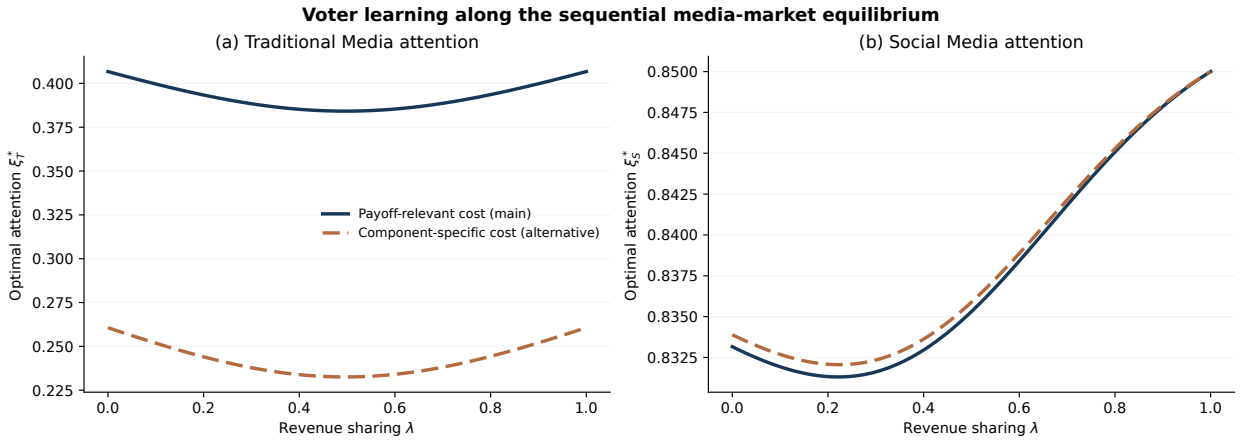


Figure 23: Optimal attention across media outlets under both mutual information costs alternatives

The ultimate effects on polarization can be seen in both figures 24 and 25. The first of them shows that as we change the revenue share parameter, our expected polarization measure follows a concave behavior of λ. Particularly, for relatively low levels of λ, expected polarization decreases on the revenue shares from Social Media that Traditional Media gets.

After some point, polarization increases. Both our benchmark and the alternative cost display the same equilibrium behavior, while the alternative cost delivers lower levels of polarization than our benchmark. The joint effects of ψ and ω appear in figure 25. The reason why there is no within-Traditional Media-polarization for the alternative is that we are showing the same value for policy stakes as in the main text. For that values (0.4) we know from 21 that there is no costly learning for Traditional Media voters. Thus, their final belief distribution is a degenerate one around their common interim mean. If we were to increase political stakes to a value that allows for positive costly learning in the alternative, our qualitative results do not change, as polarization keeps being decreasing on ψ.

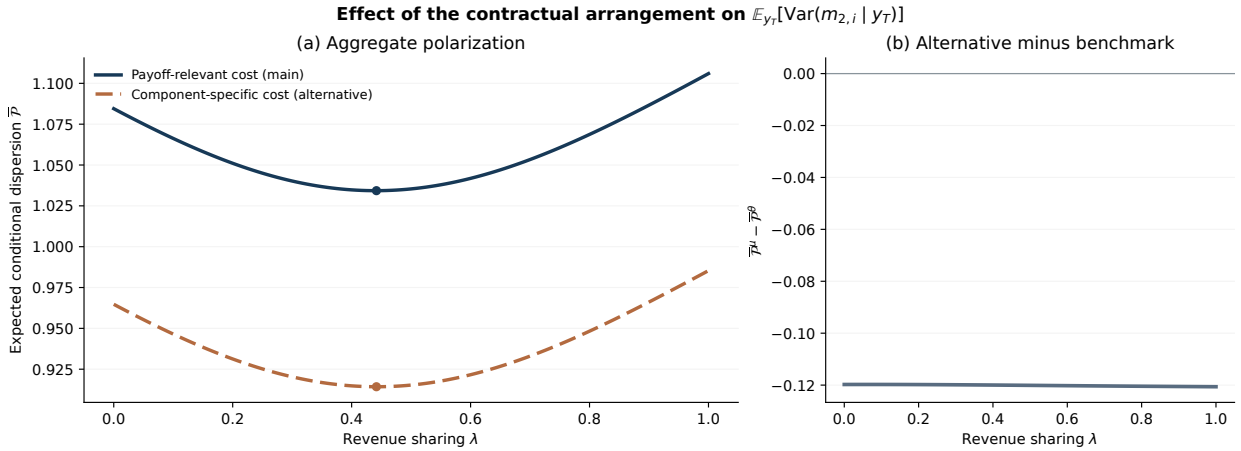


Figure 24: Effects of λ on expected polarization. Two mutual information costs.

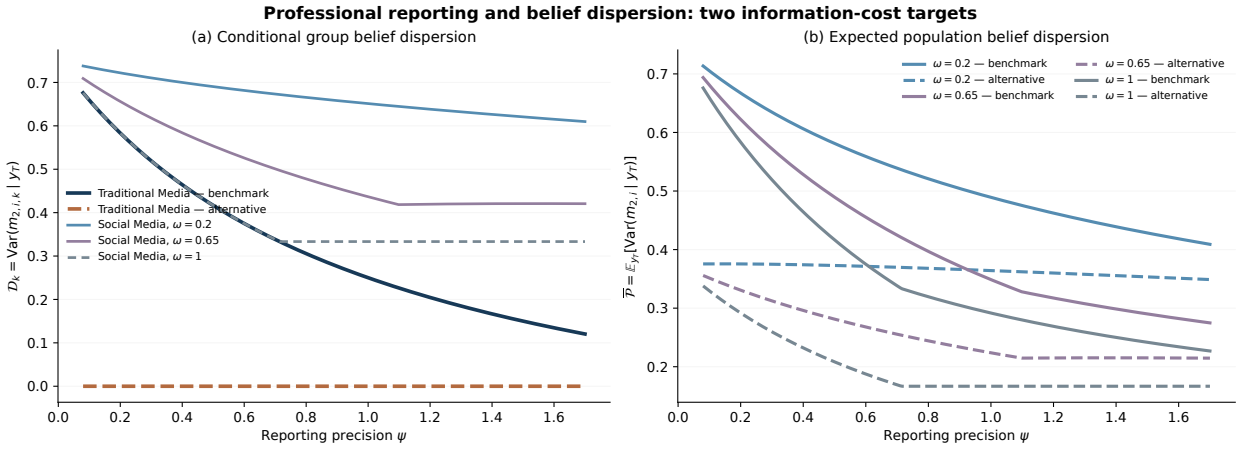


Figure 25: Effect of optimal precision on polarization. The figures keep ω constant.